UNITY GAME TEMPLATE BLOCKY HOLES

USER GUIDE

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I. TEMPLATE INTRODUCTION

Blocky Holes is an exciting endless level-base game in which you touch the screen to control the flaming hole and moving around on the surface, avoid the black objects, collect all other objects to recieve cash as reward and use it to unlock new holes and complete the level. The game is ready to release straight out of the box, and it can also be easily customized to make it even more engaging to your players. Supports for PC/Mac, iOS, Android, etc…!

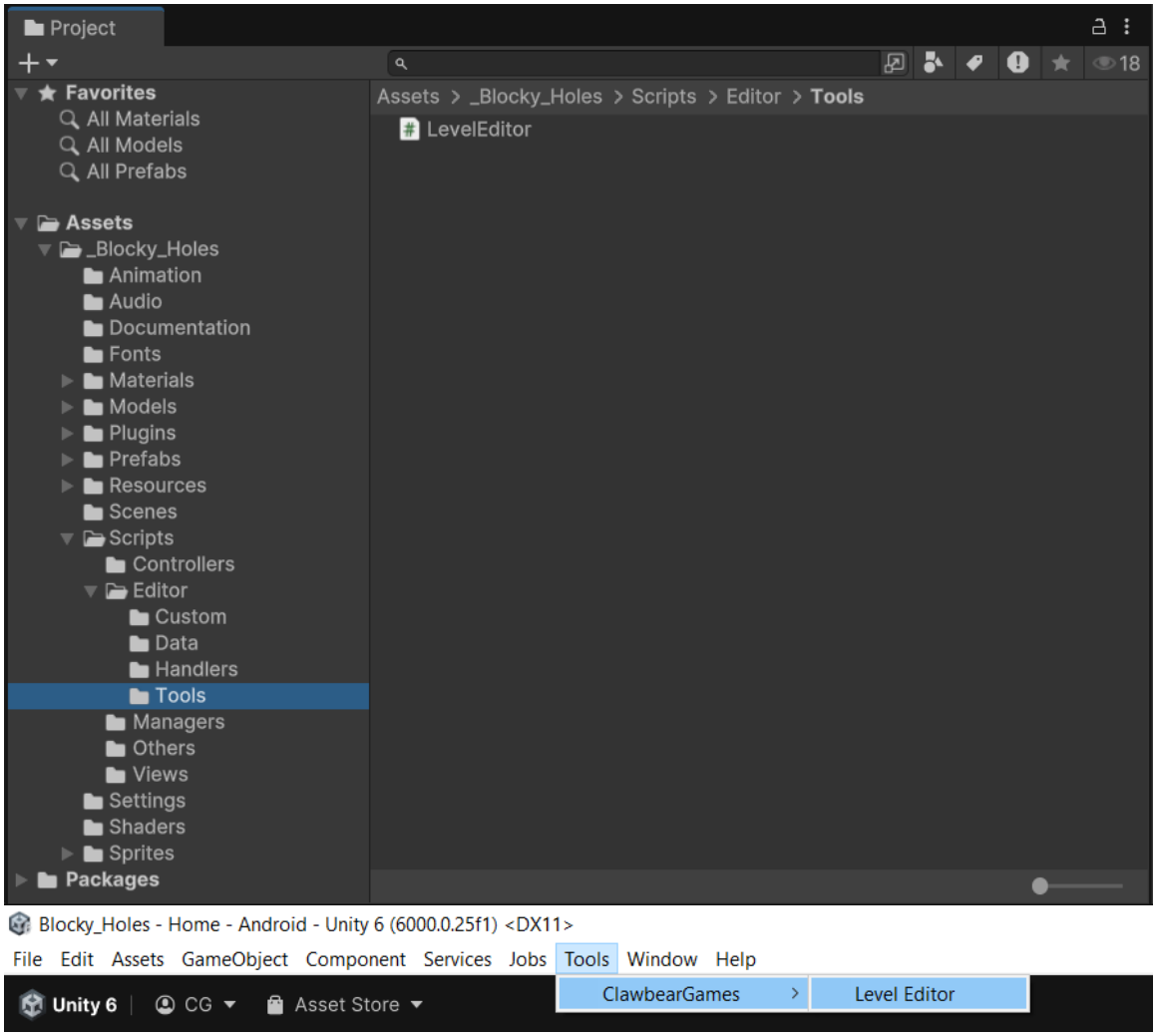
This template is made with Unity C# and optimized for mobile devices. This template provides you a full-featured, ready-for-release game source code that you can customize and build your own game in no time!

Highlight features:

- Addictive one-touch gameplay
- Smooth control, eye-catching blocky graphics
- 50++ levels (more level will be added soon)
- Very easy to add new levels with detailed instruction

- Easy to use built-in Level Editor
- Leaderboard system using Dreamlo
- Notification system for Android/iOS
- Daily reward system for better retention
- Multiple ad networks: Admob and Unity Ads ready to use (banner, interstitial and rewarded ad).
- Commented C# code with detailed documentation
- Optimized for mobile
- Free-to-use assets (fonts, sounds, music, models, etc.)
- Ready to publish out-of-the-box

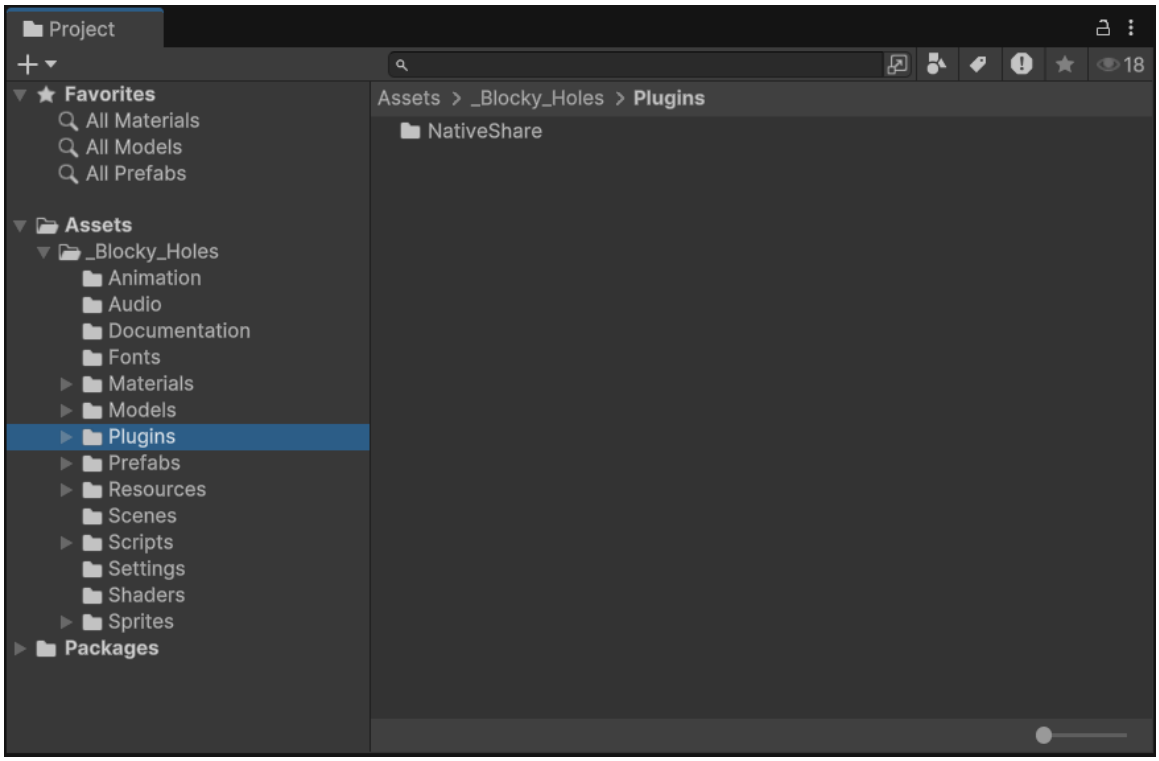
This template contains an editor extension used to design level. Editor extension file is located at: Assets/_Blocky_Holes/Scripts/Editor/Tools and the editor tool is placed under Tools/ClawbearGames/.

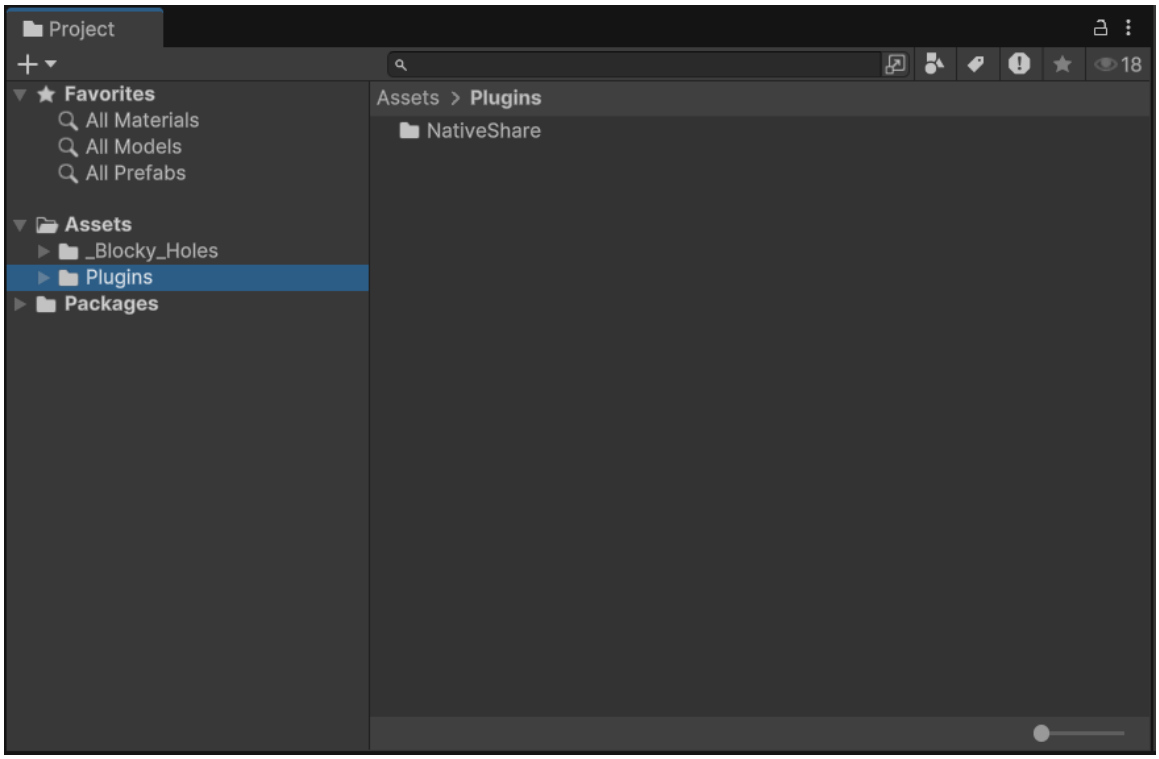


II. TEMPLATE SETUP

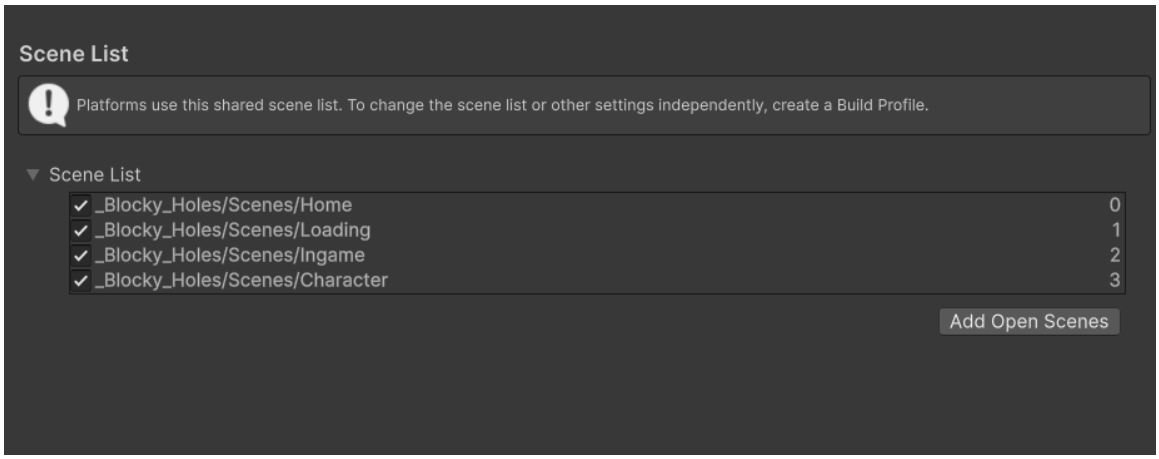
This template was designed for mobile (Android, iOS, Windows Phone…) so after imported the package to unity, you need to switch to Android or iOS, or Window Phone.

After that, go to Assets/_Blocky_Holes and move the folder Plugins out of _Blocky_Hole folder.





The template contains 4 scenes: Home, Loading, Fighter and Ingame under the path Assets/_Blocky_Holes/Scenes. **You need to start from Home scene first. Do not start from Loading, Fighter or Ingame scene.**

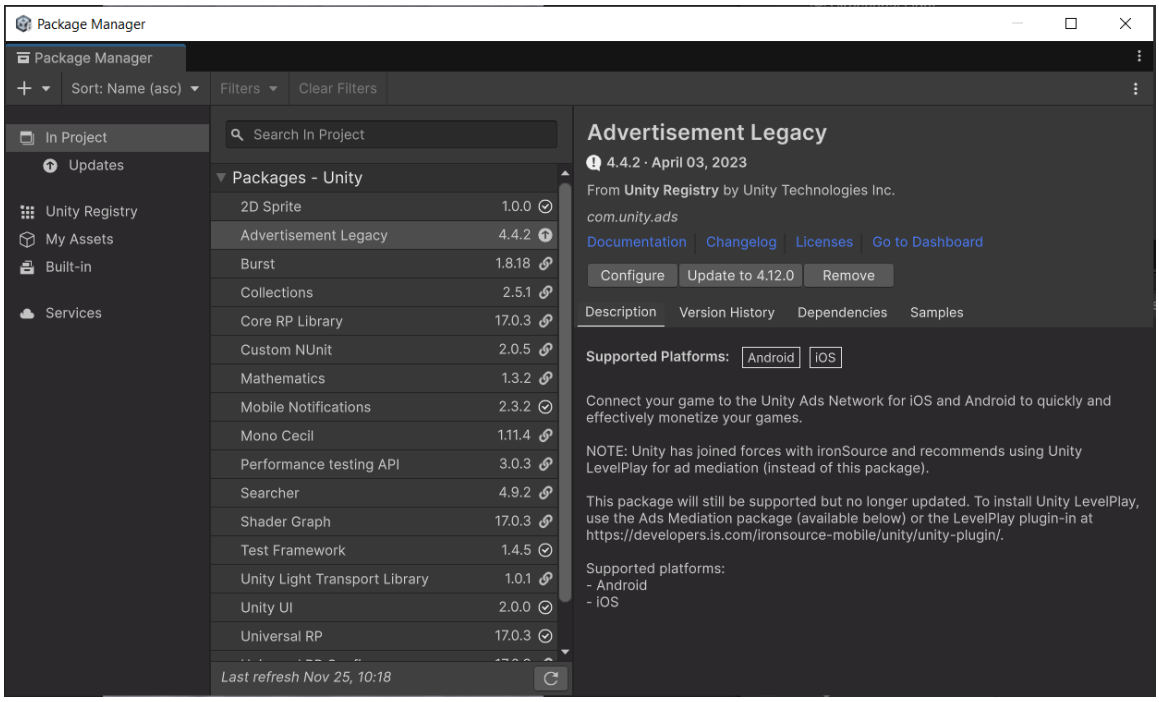


III. REQUIREMENT PACKAGES

When you open this template, at first you will some errors in Console

window, that's because this template requires some packages to run. You have to install these packages bellow to have this template run smoothly. Please follow these instructions:

- Open Package Manager by go to Window/Package Manager. If you don't see Package Manager button, then you have to re-install your unity completely because it's the problem that Unity Technologies still not able to fix.

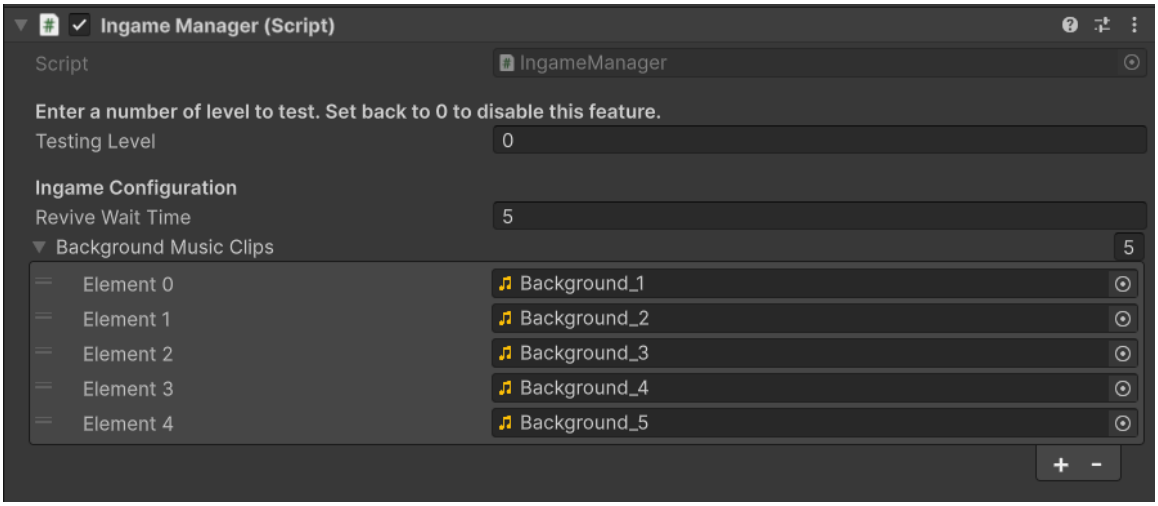


- These are packages you must need to install:
  - Advertisement version 4.4.2 or higher.
  - AI Navigation version 2.0.4 or higher
  - Mobile Notification version 2.3.2 or higher
  - 2D Sprite version 1.0.0 or higher
  - Unity UI 1.0.0 or higher
  - Universal RP version 17.0.3 or higher

IV. GAMEPLAY CUSTOMIZATION

1. Ingame Manager

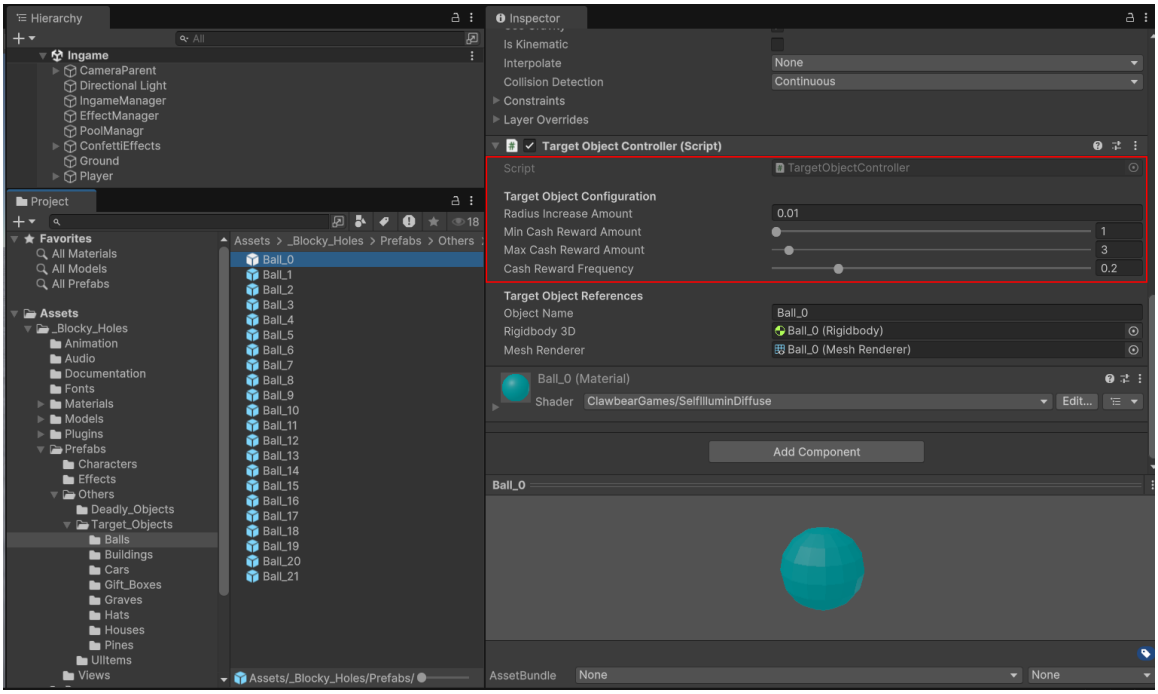
Most of important gameplay parameters can be configured within the IngameManager component which is attached to a game object also named IngameManager in the hierarchy. You can find IngameManager object in InGame scene.



- *Testing Level:* the level you need to test before building the game. If you set it to 0, the script will not load the testing level.
- *Revive Wait Time:* the delay time for revive feature last.
- *Background Clips:* the list of background songs that play randomly when a level started.

2. Target Object Controller

The target object is the object that player will collect to complete the level. Its parameters can be configured within the TargetObjectController component which is attached to all the target objects. You ca find all target object at Assets/_Blocky_Holes/Prefabs/Others/Target_Objects.



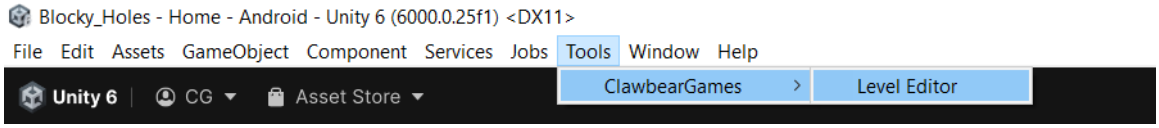
- *Radius Increase Amount:* the amount that the radius of the player's

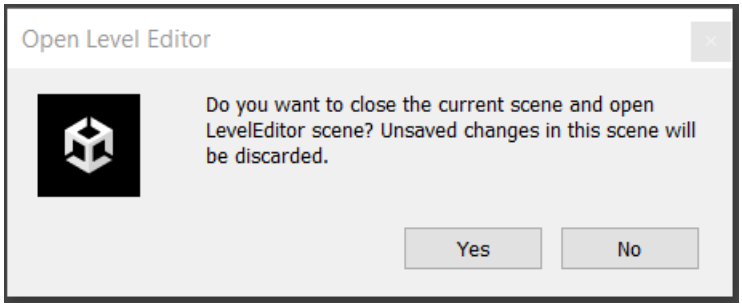
hole will increase after collecting this target object.

- *Min Cash Reward Amount:* the minimum amount of cash reward the layer will get.
- *Max Cash Reward Amount:* the maximum amount of cash reward the layer will get.
- *Cash Reward Frequency:* the frequency to create cash reward.

V. LEVEL CUSTOMIZATION

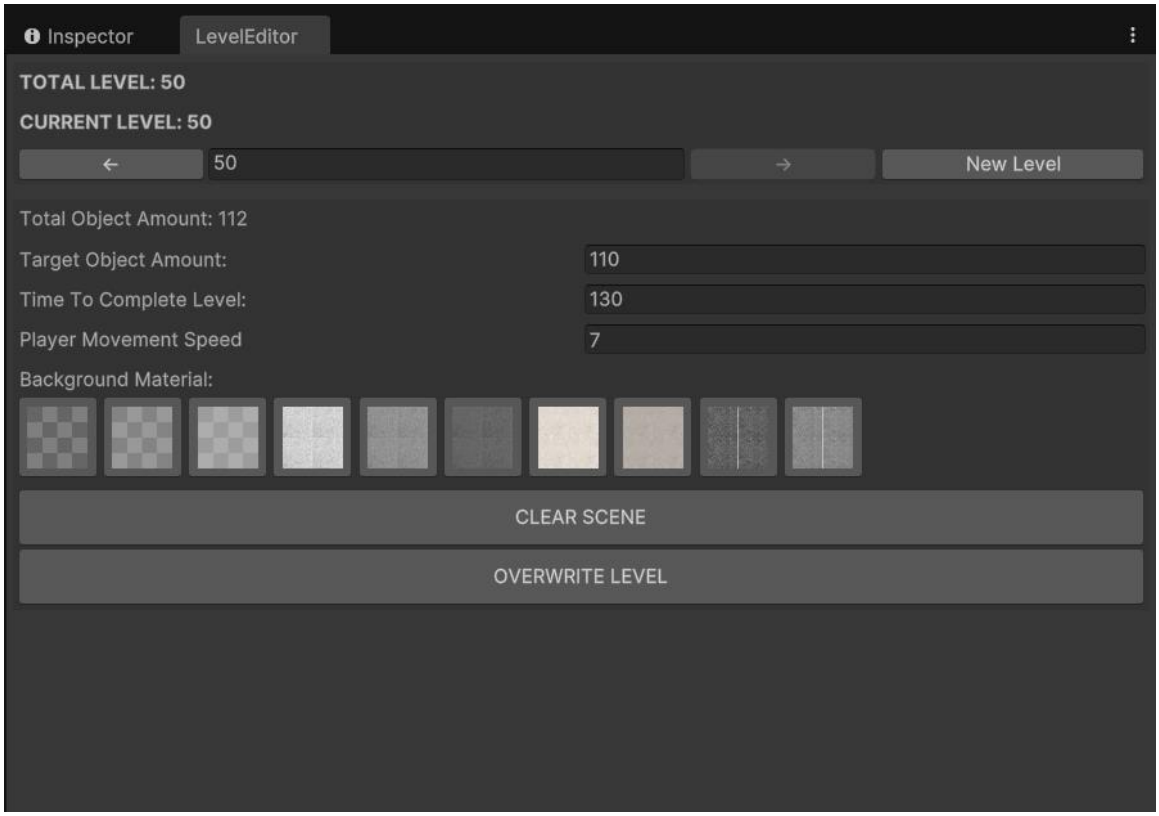
This is a level-base game template and the goal of the game is you have to complete a level and start a new level. In this section I will show you how to use the level editor to create new level and edit existing level. To open the level editor go to *Menu Items/Tools/Level Editor*. Remember to save the scene before open the level editor, unsaved changes in this scene will be discarded.





1. Create New Levels

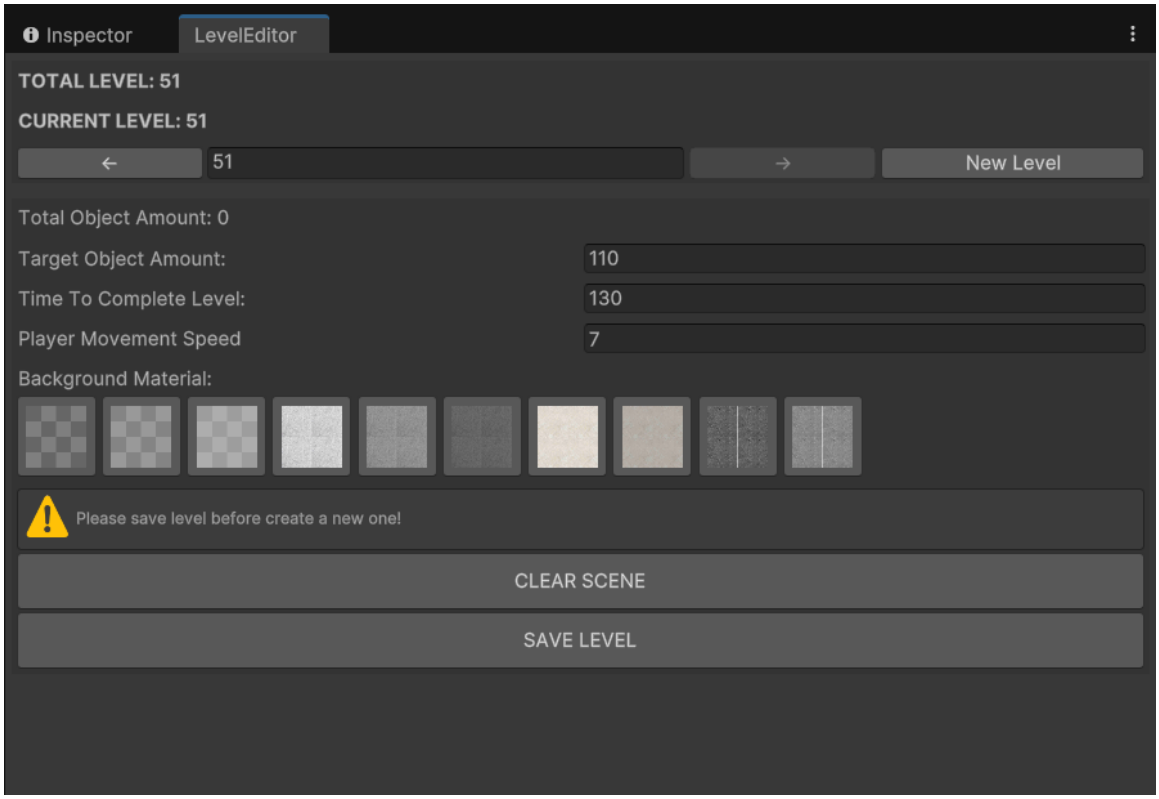
To open the level editor go to *Menu Items/ClawbearGames/Tools/Level Editor* and you will be direct to LevelEditor scene and an editor window like this will show up.



- TOTAL LEVEL: the total level number of the project.
- CURRENT LEVEL: the current level number.
- " ← " button: load previous level button.
- " → " button: load next level button.
- New Level button: create new level button.
- Total Object Amount: the total target object in the level.
- Target Object Amount: the amount of target object player needs to collect to complete the level.
- Time To Complete Level: the time to complete the level.
- Player Movement Speed: the movement speed of the player.
- Background Material: the material of the background.
- CLEAR SCENE: clear all the black dots and colored dot in the scene.
- OVERWRITE LEVEL: convert the level to json string and save it to json file in the *Resources/Levels* folder path.

How to create new level:

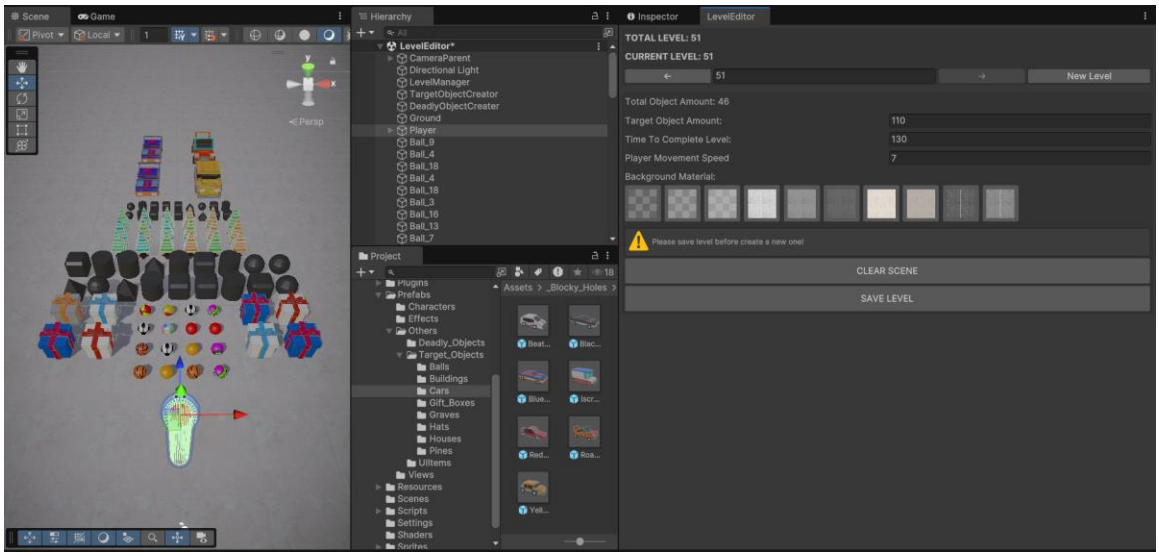
Step 1: click New Level button.



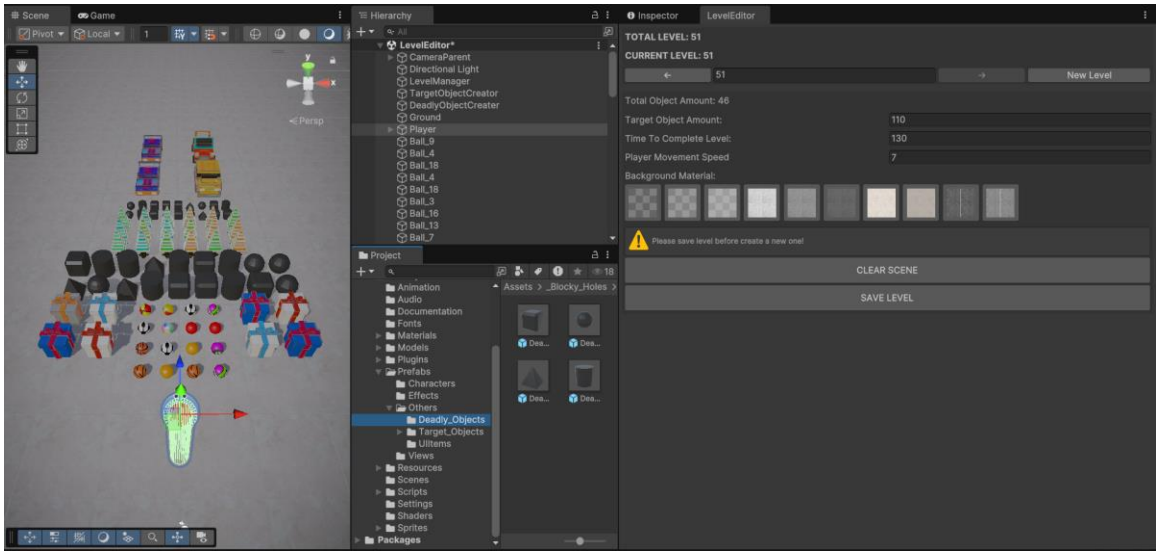
There is a message tell you that you need to save the level before create a new one. That mean you just create a json file for the new level and that json file has no data. You need to save the data for that json file.

Now click CLEAR SCENE button to clear the scene. All the target objects and deadly objects will be destroyed. **YOU CAN MODITY THOSE TARGET OBJECTS AND DEADLY OBJECTS AS YOU WANT FOR THE NEW LEVEL TO SAVE YOUR TIME.** Now I just want to design a new level that difference with all the level I have, so I click CLEAR SCENE.

Now, go to Assets/Prefabs/Others/Target_Objects and drag the target object prefabs that you want into the Hierarchy, adjust the position, the rotation and the scale as you want.

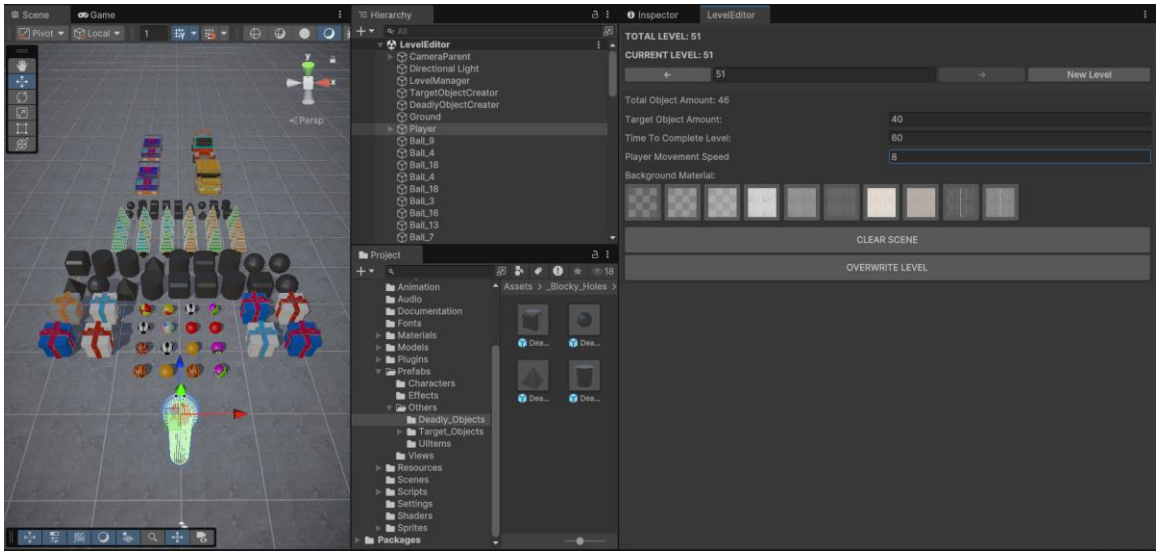


Now when you done with the target objects, go to Assets/Prefabs/Others/Deadly_Objects and drag the deadly objects you want into the Hierarchy, adjust the position, the rotation and the scale as you want.



PLEASE KEEP IN MIND THAT A LEVEL NEED AT LEST 1 TARGET OBJECT PREFAB AND 1 DEADLY OBJECT PREFAB, OTHERWISE, YOU NOT BE ABLE TO SAVE THE LEVEL.

Now adjust the Target Object Amount to be lower than the Total Object Amount, adjust the Time To Complete Level, Player Movement Speed and Background Material as you want and click SAVE LEVEL to save the level.

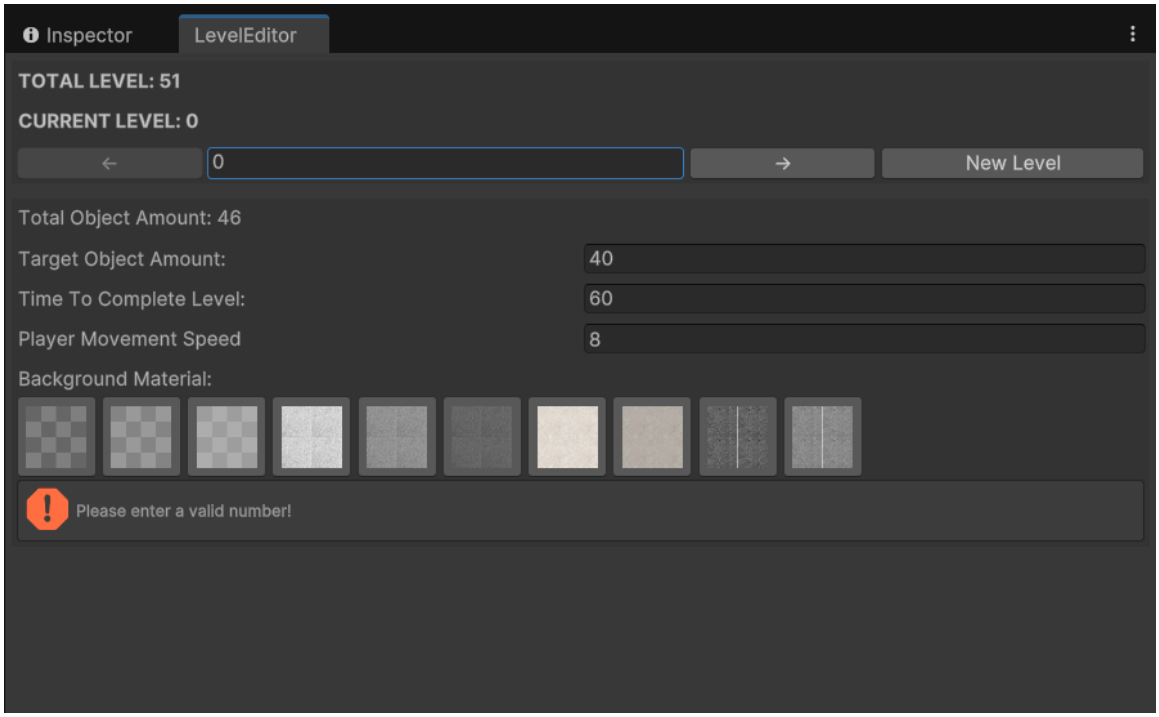


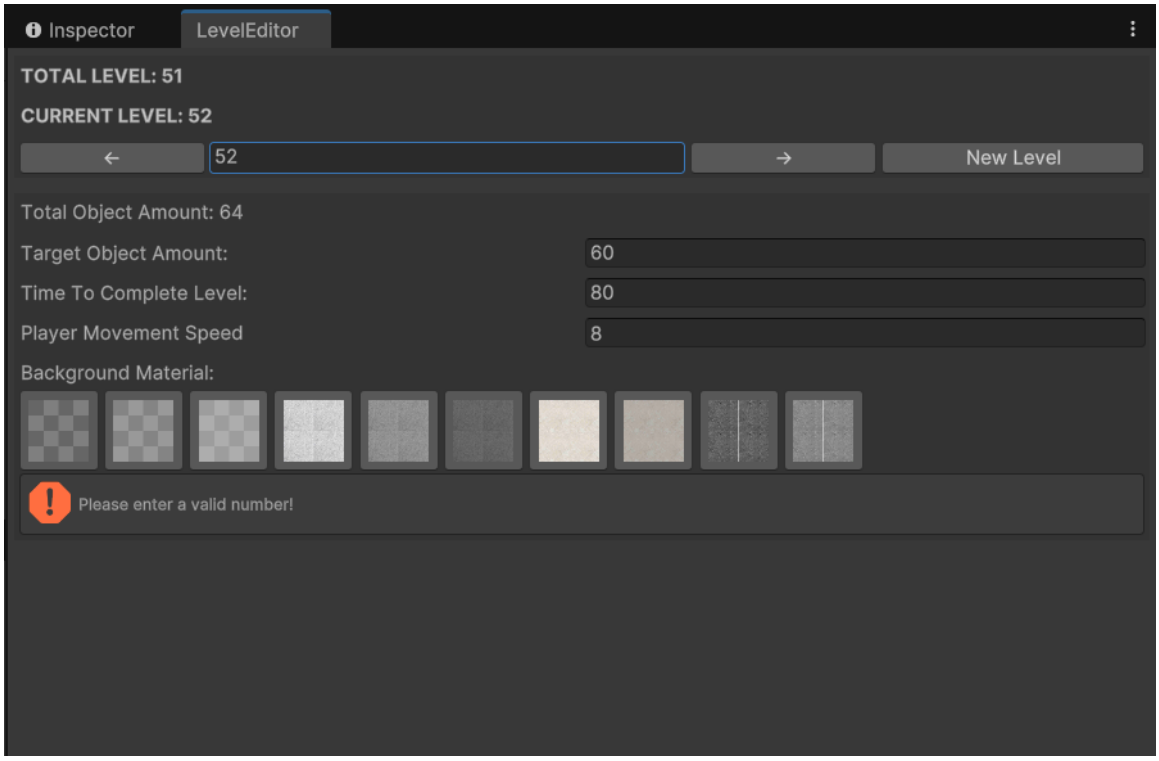
Ok, we done with create new level and save it.

2. Edit Existing Levels

To edit existing level, use “ ← ” button and “ → ” button to load the level you want to edit, or just type the number in Current Level field to load the level.

NOTE: DO NOT ENTER THE NUMBER THAT SMALLER OR EQUAL THAN 0 AND LARGER THAN THE TOTAL LEVEL OR YOU WILL GET THIS ERROR MESSAGE.





After loaded the level you want to edit, you can click CLEAR SCENE to destroy all the target objects and deadly objects, then drag and drop new objects into the Hierarchy to design new level. Or just simply adjust the parameters of target objects and deadly objects currently in the Hierarchy.

After you done with the new design of existing level, just hit OVERWRITE

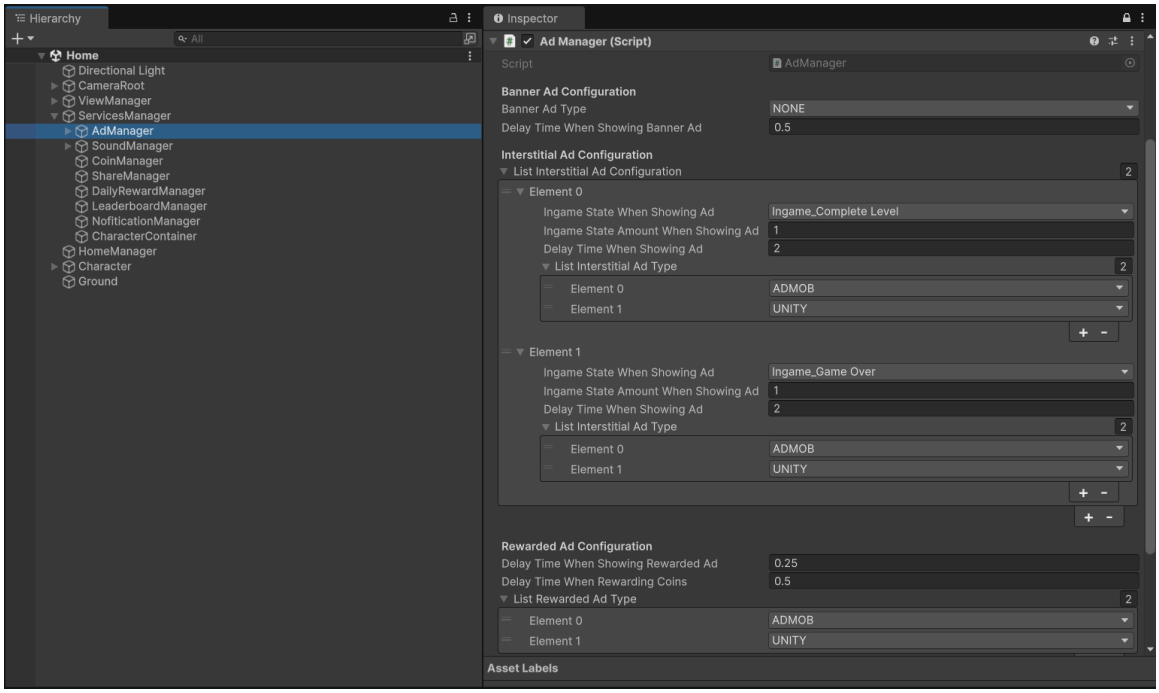
LEVEL. That's it.

VI. SERVICES CONFIGURATION

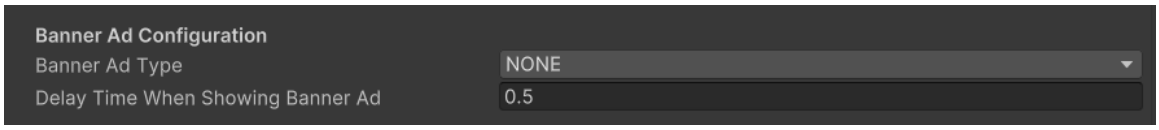
This template comes with many features like multiple ads, leaderboard, daily reward, notification…ect. You can find all the scripts for these features under ServicesManager object in Home scene.

1. Ad Manager

The AdManager object in hierarchy of scene Home contains AdManager component, in which you can customize which type of ads you want to use and how you want to show the ads.



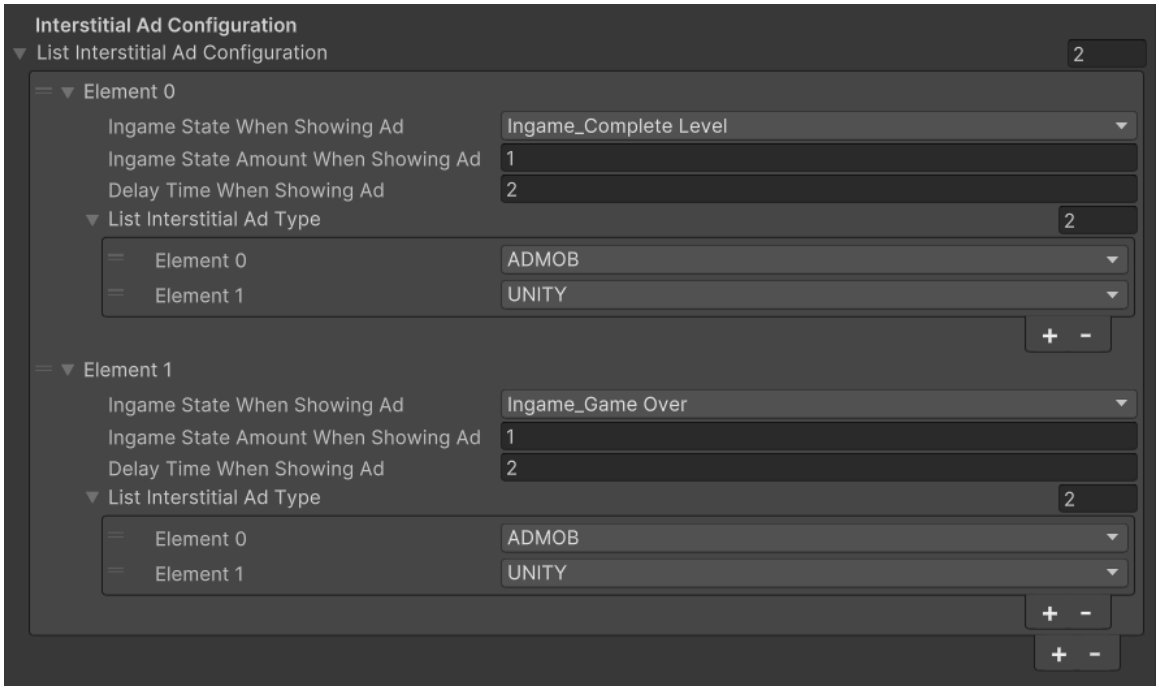
Banner Ad Configuration: this is the section where you can control which type of banner ad you want to show. Currently, the template support for 2 types of banner ad: Admob and Unity.



- *Banner Ad Type:* the type of banner ad you want to show.

- *Delay Time When Showing Banner Ad:* the delay time when showing banner ad.

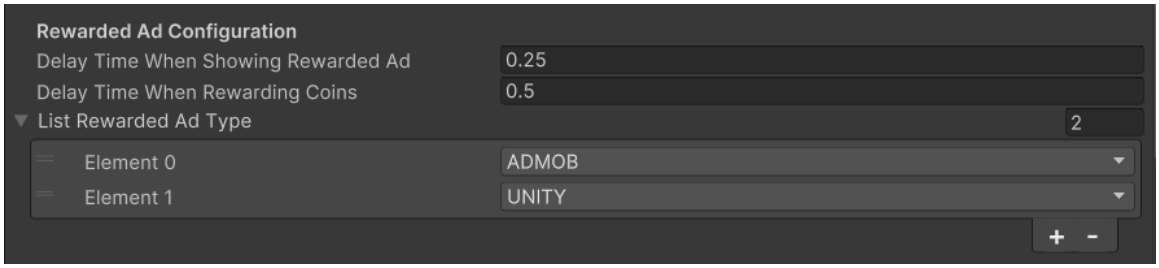
Interstitial Ad Configuration: this is the section where you can control which type of interstitial ad you want to show and how you want to show it. Currently, the template support for 2 types of interstitial ad: Admob and Unity.



- *Ingame State When Showing Ad:* the game state you want to show the ad.
- *Ingame State Amount When Showing Ad:* the amount of game state that player go through to show ad. Example: if the value is 2 and *Ingame State When Show Ad* is Ingame_GameOver, that mean the ad will show after 2 times of game over.
- *Delay Time When Showing Ad:* the delay time when showing the interstitial ad.
- *List Interstitial Ad Type:* the list of interstitial ad type. The first item of this list will be the first priority of interstitial ad. Example: in the image you can see 2 items: ADMOB and UNITY. ADMOB is the first item so, *Ad Manager* will load the interstitial ad of Admob first, if there's no Admob's interstitial ad to load, then *Ad Manger* will continue to load Unity interstitial ad.

Rewarded Ad Configuration: this is the section where you can control

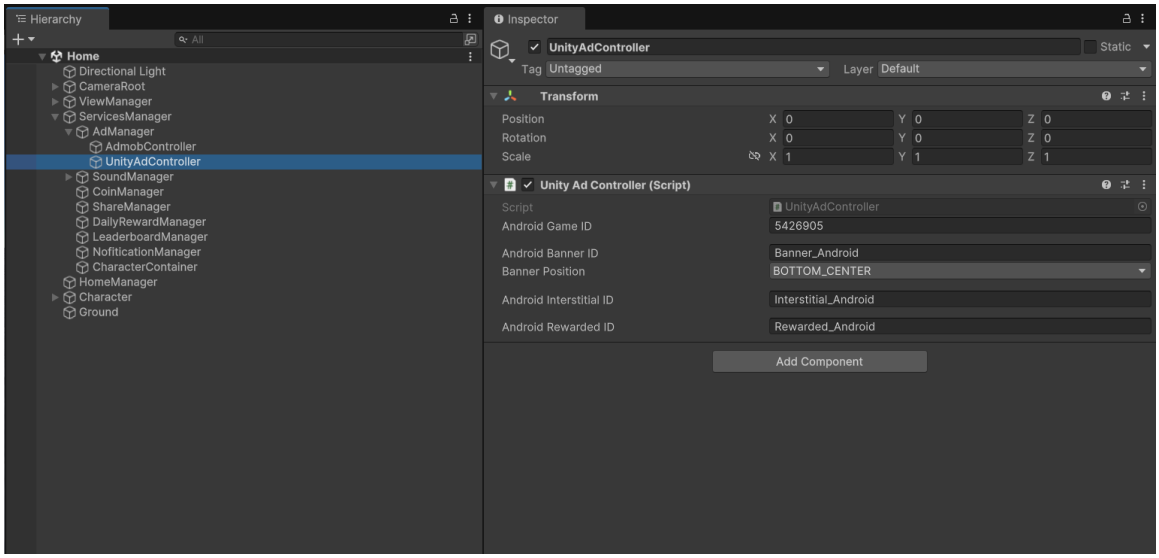
which type of rewarded ad you want to show and how you want to show it. Currently, the template support for 2 types of rewarded ad: Admob and Unity.



- *Delay Time When Showing Rewarded Ad:* the delay time when showing rewarded ad.
- *Delay Time When Rewarding Coins:* the delay time when rewarding coins to user after they watched rewarded ad.
- *List Rewarded Ad Type:* the list of rewarded ad type. The first item of this list will be the first priority of rewarded ad. Example: in the image you can see 2 items: ADMOB and UNITY. ADMOB is the first item so, *Ad Manager* will load the rewarded ad of Admob first, if there's no Admob's rewarded ad to load, then *Ad Manger* will continue to load Unity rewarded ad.

2. Unity Ad Controller

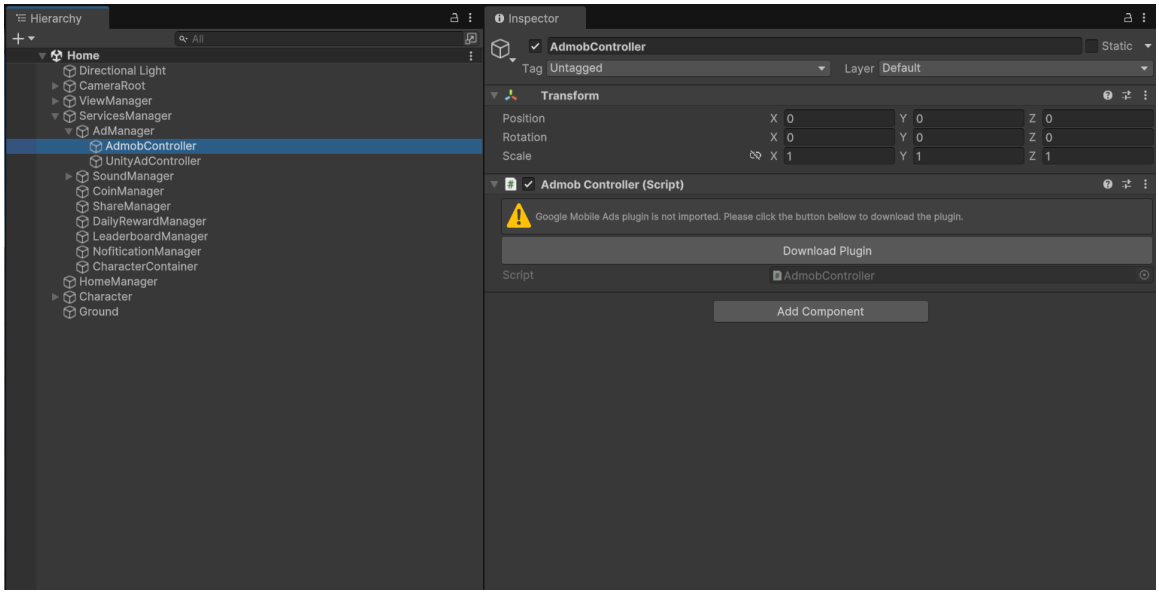
Before setting up Unity ad, you need to set up your project for Unity Services first, you can see the instruction [here](#). After setting up Unity Services, just put your game id and your ad ids and you will good to go.



- *Android Game ID:* the android id of your unity ad project (it will show the id for iOS when you switch to iOS). You can find all of your ad projects here.
- *Android Banner ID:* the android banner id of your ad project (it will show the id for iOS when you switch to iOS).
- *Banner Position:* the position of the banner ad.
- *Android Interstitial ID:* the interstitial id of your ad project (it will show the id for iOS when you switch to iOS).
- *Android Rewarded ID:* the rewarded ad id of your ad project (it will show the id for iOS when you switch to iOS).

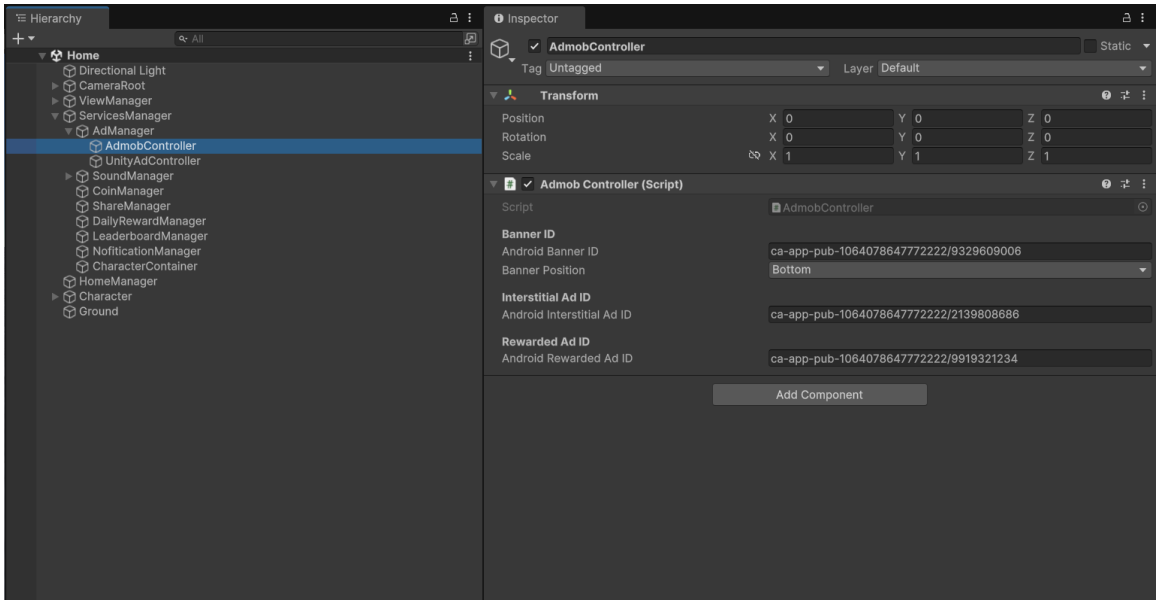
3. Admob Controller

The AdmobManager object in hierarchy of scene Home contains AdmobController component, in which you can customize parameters like admob id, ad units...



As you can see in the image, there's no option to change ads units because the template did not have google mobile ads plugin included, so if you want to use Admob, please click to the Download Plugin button, download the latest version of google mobile ads and import it into the project.

After imported the plugin, select AdManager object in the hierarchy, wait for few seconds for the project rebuild and you will see this.

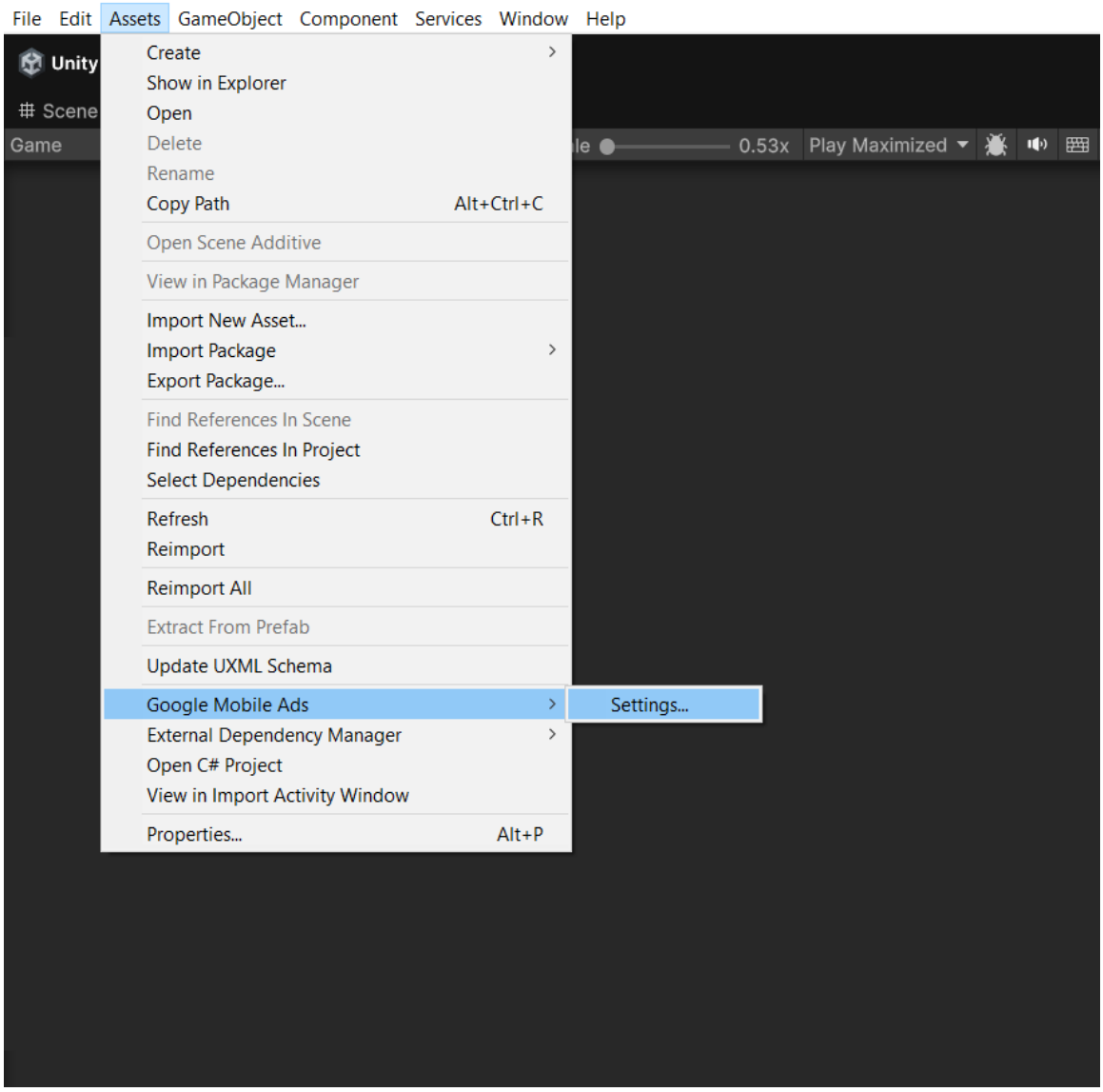


Now you can config Admob id and all the ad units as you want. Currently, the platform using is Android, that why all the ids in the image showing for Android, of course it will show the ids for iOS when you switch the build platform to iOS.

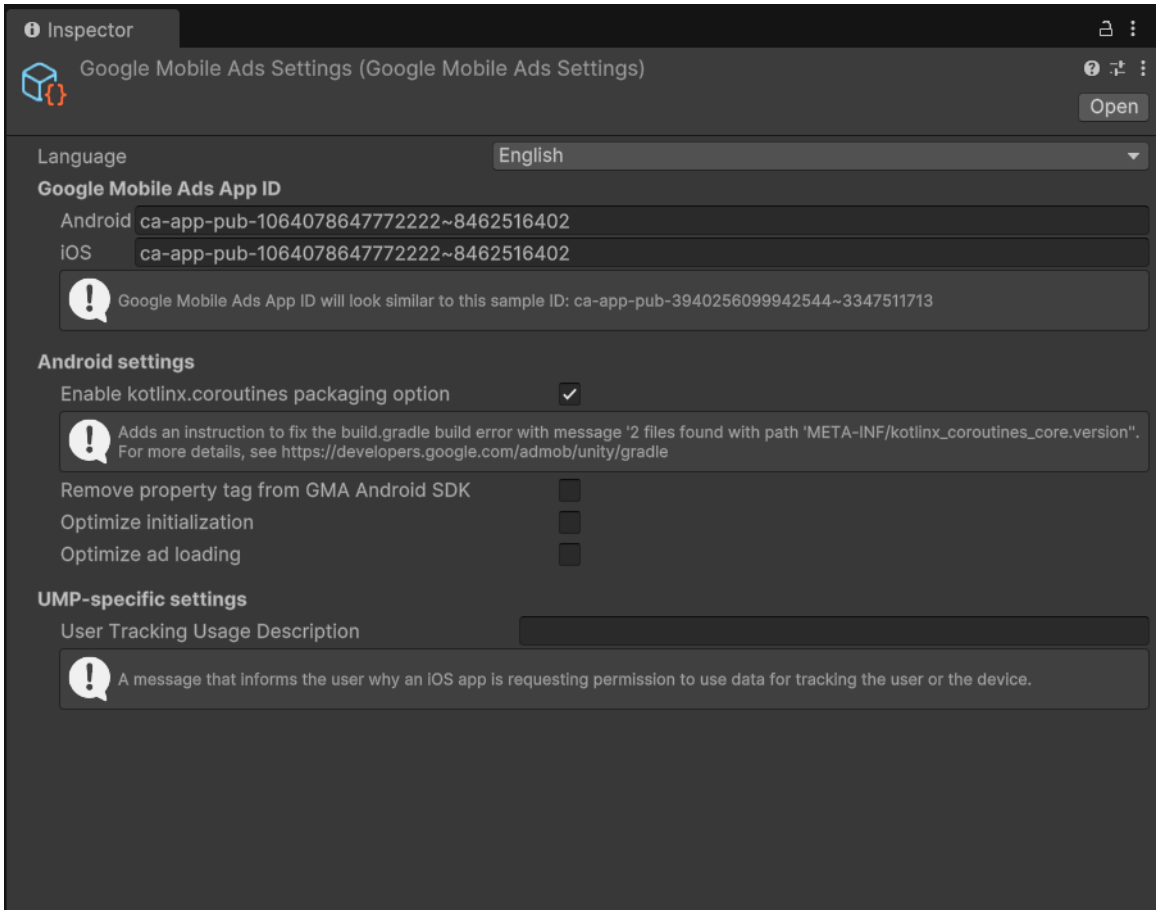
- *Android Banner ID:* the id of banner ad in your Admob account (it will show the id for iOS when you switch to iOS).
- *Banner Position:* the position of the banner ad.
- *Android Interstitial ID:* the id of interstitial ad in your Admob account (it will show the id for iOS when you switch to iOS).
- *Android Rewarded ID:* the id of rewarded ad in your Admob account (it will show the id for iOS when you switch to iOS).

Now we need to setup the Admob App ID. Please follow these steps to setup the Abmob App ID.

- Go to Assets/Google Mobile Ads/Setting



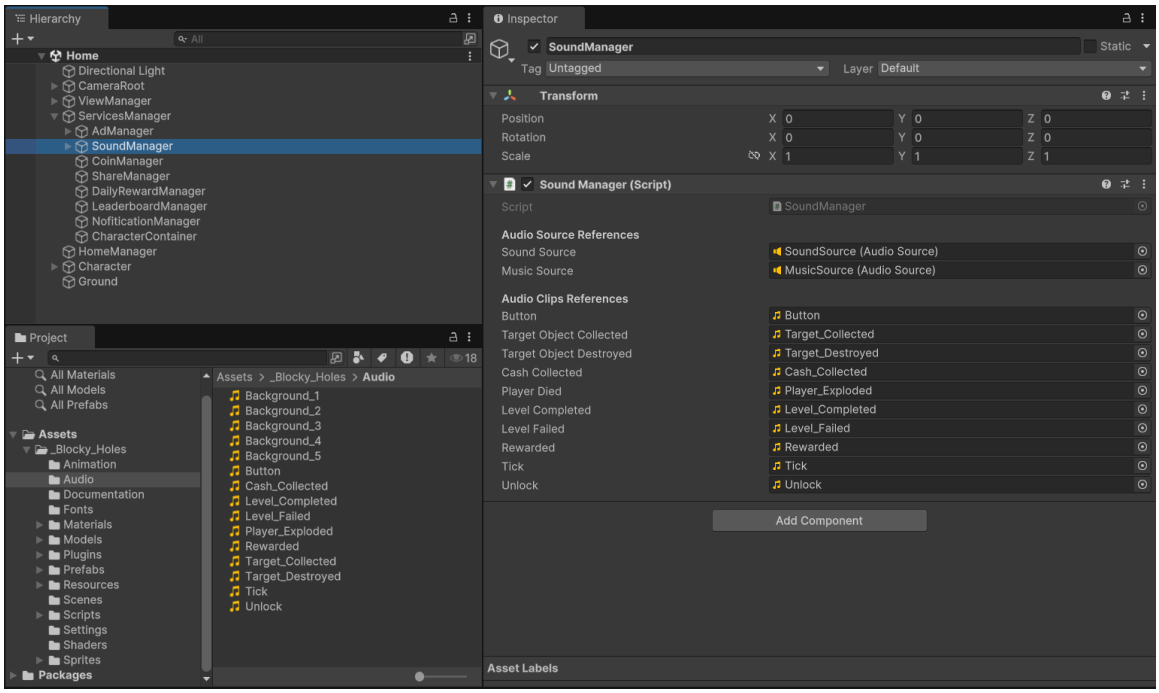
- Insert the Google Mobile Ads App ID at Android/iOS field base on your current platform.



- Hit Ctrl + S to save, that's all.

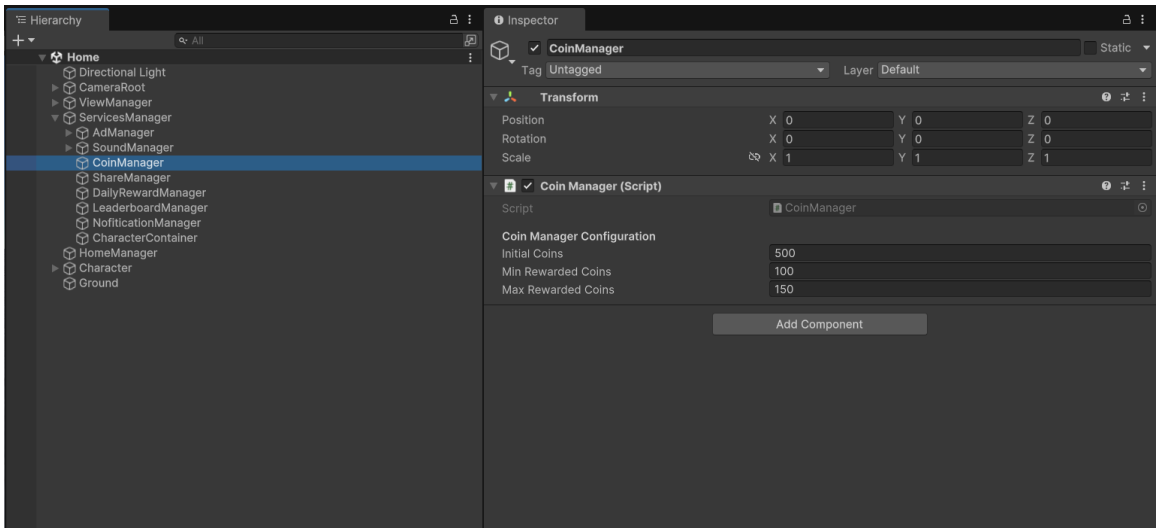
4. Sound Manager

All sounds included in this game are free-to-use in commercial projects and are located under the path *Assets/_Blocky_Holes/Audio* folder.



5. Coin Manager

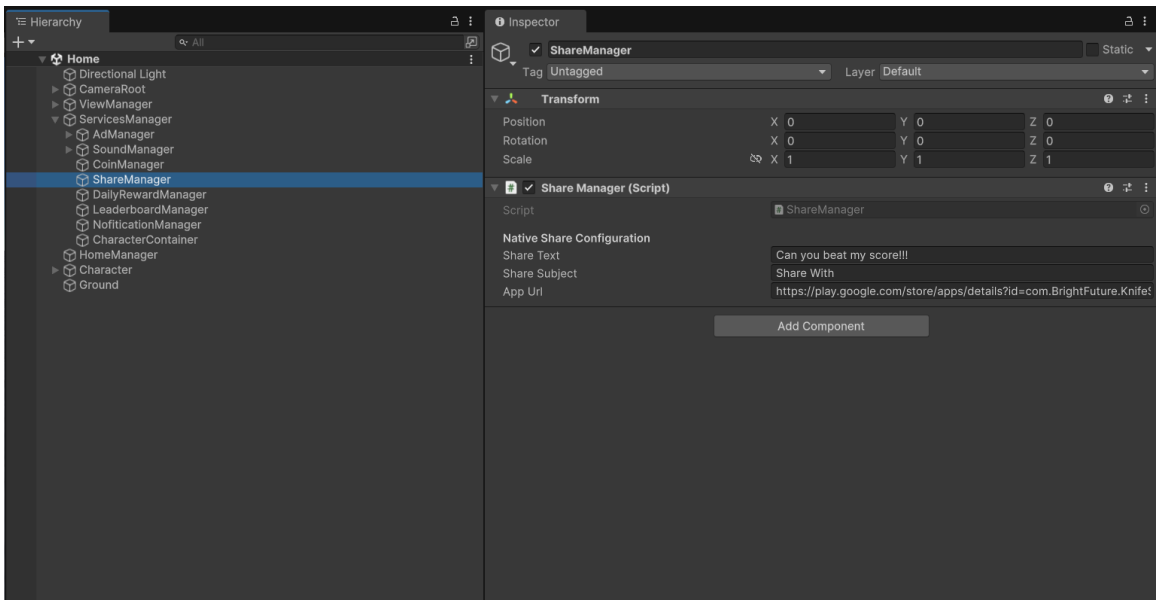
Coin is the currency in the game that player collect to unlock new characters. All information about coin can be config in CoinManager game object.



- *Initial Coins:* the initial coins that player will have when the game start at the first time.
- *Min Rewarded Coins:* the minimum coin amount that user will be rewarded after they watched rewarded ad.
- *Max Rewarded Coins:* the maximum coin amount that user will be rewarded after they watched rewarded ad.

6. Share Manager

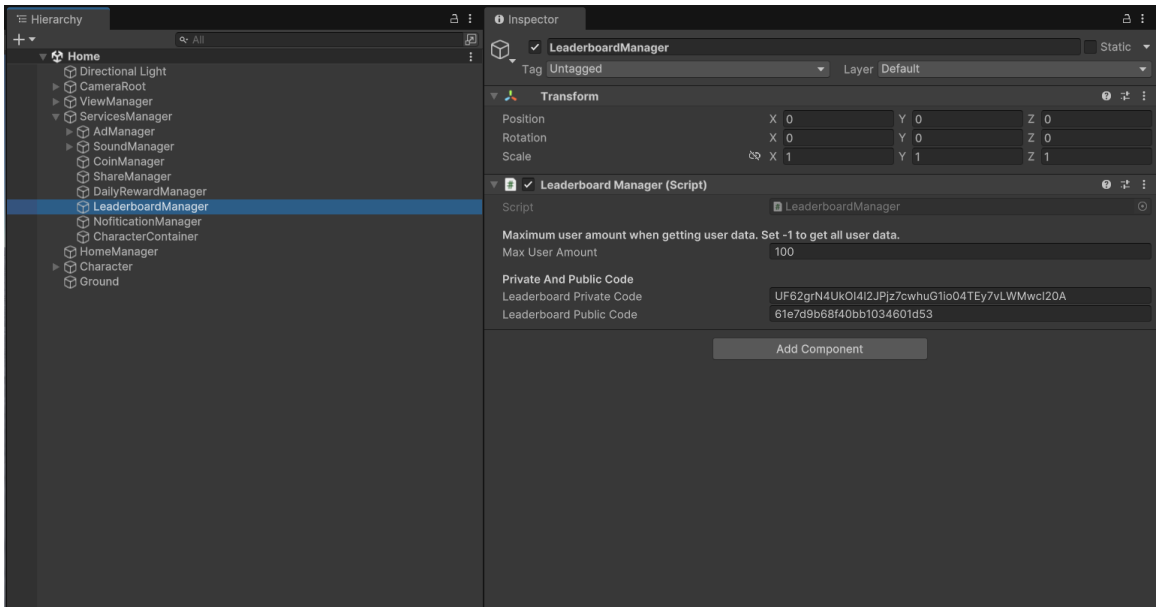
All information for sharing feature can be config in ShareManager game object. It contains the information likes shreenshot's name, text, subject and url…You can config these features from the ShareManager object in the hierarchy.



- *Share Text:* the text for sharing feature.
- *Share Subject:* the subject for sharing feature.
- *App Url:* the url of the app (Google Play on Android and App Store on iOS).

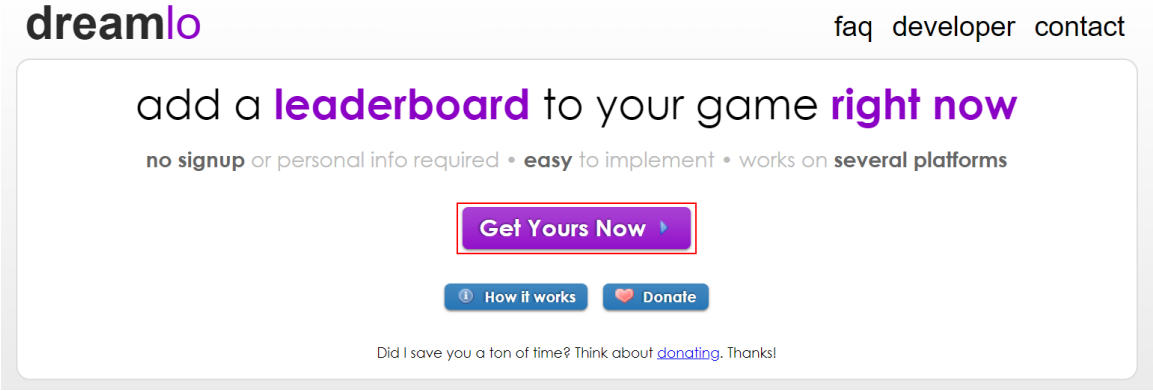
7. Leaderboard Manager

All information for leaderboard feature can be config in LeaderboardManager game object. It contains 2 parameters for you to config the leaderboard. Currently, we used Dreamlo to handle leaderboard feature.

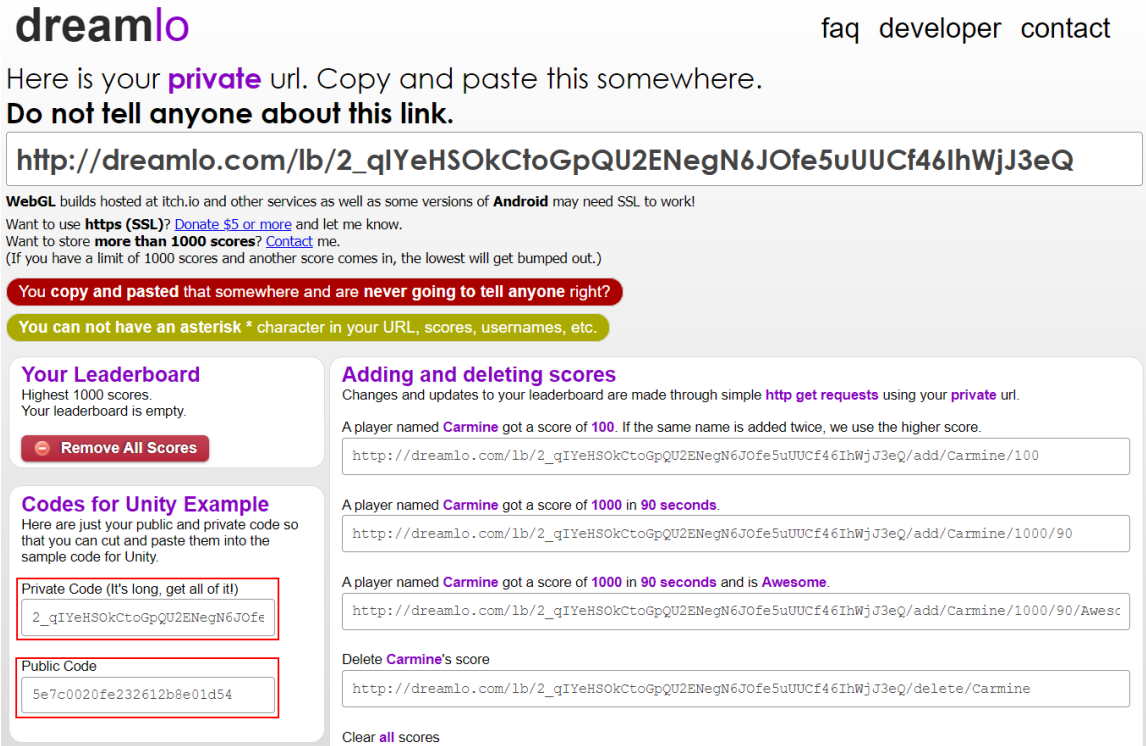


- *Max User Amount:* the maximum amount user show in the leaderboard. If you set it to -1, it will show all the user that played the game and report to leaderboard table. We suggest you to limit it to 50 or 100, because if your game have 1000 downloads, which mean you have 1000 user in the leaderboard and when you load it all to the leaderboard view, the game will become laggy.
- *Leaderboard Private Code:* the leaderboard private code of Dreamlo.
- *Leaderboard Public Code:* the leaderboard public code of Dreamlo.

Now we will show you how to get the private and public code from Dreamlo. First, you need to go to Dreamlo.com, at the top of the front page, click to Get Yours Now button.



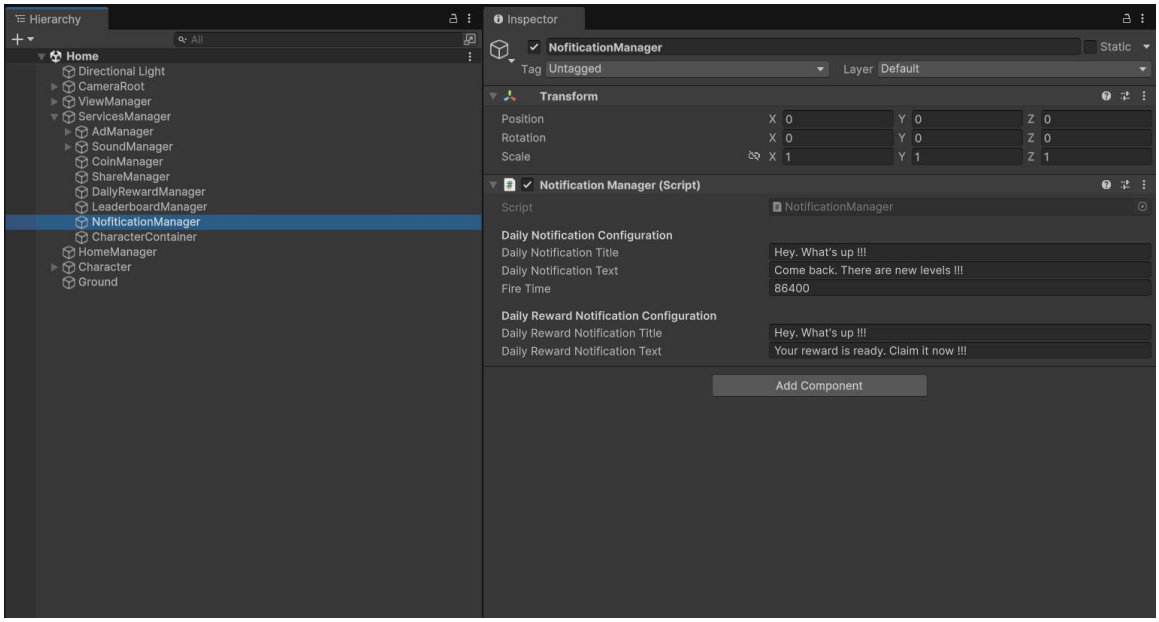
After that, you will be direct to a page that you can get your private and public code.



Copy 2 codes and paste it on LeaderboardManager component. That's it.

8. Notification Manager

All information for notification feature can be config in NotificationManager game object. It contains 3 parameters for you to config the notification. Currently, we used Mobile Notification package provided by Unity to handle the notification.



Daily Notification Configs: this is the notification that fire daily.

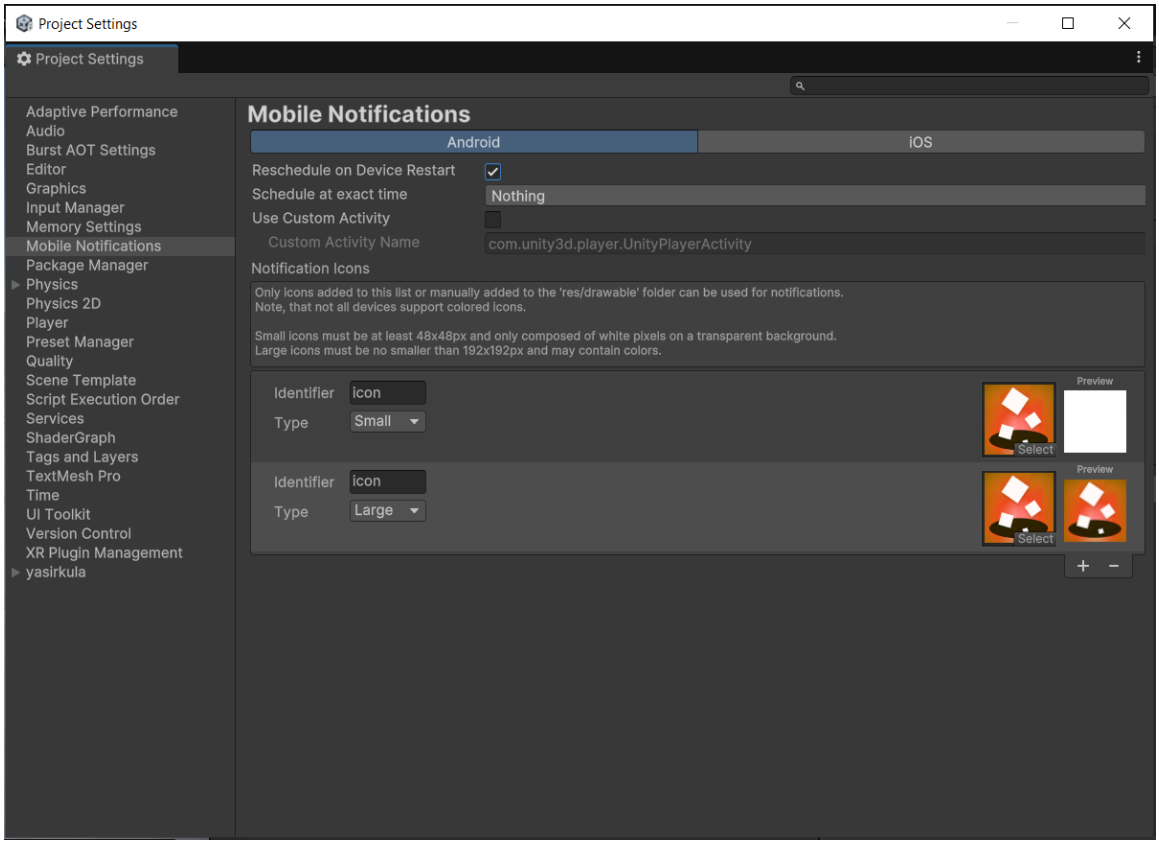
- *Daily Notification Tile:* the tile of the notification when it fired.
- *Daily Notification Text:* the text of the notification when it fired.
- *Fire Time:* how long till the notification get fire (in seconds) after the game close. Currently it's 86400 seconds which is 1 day.

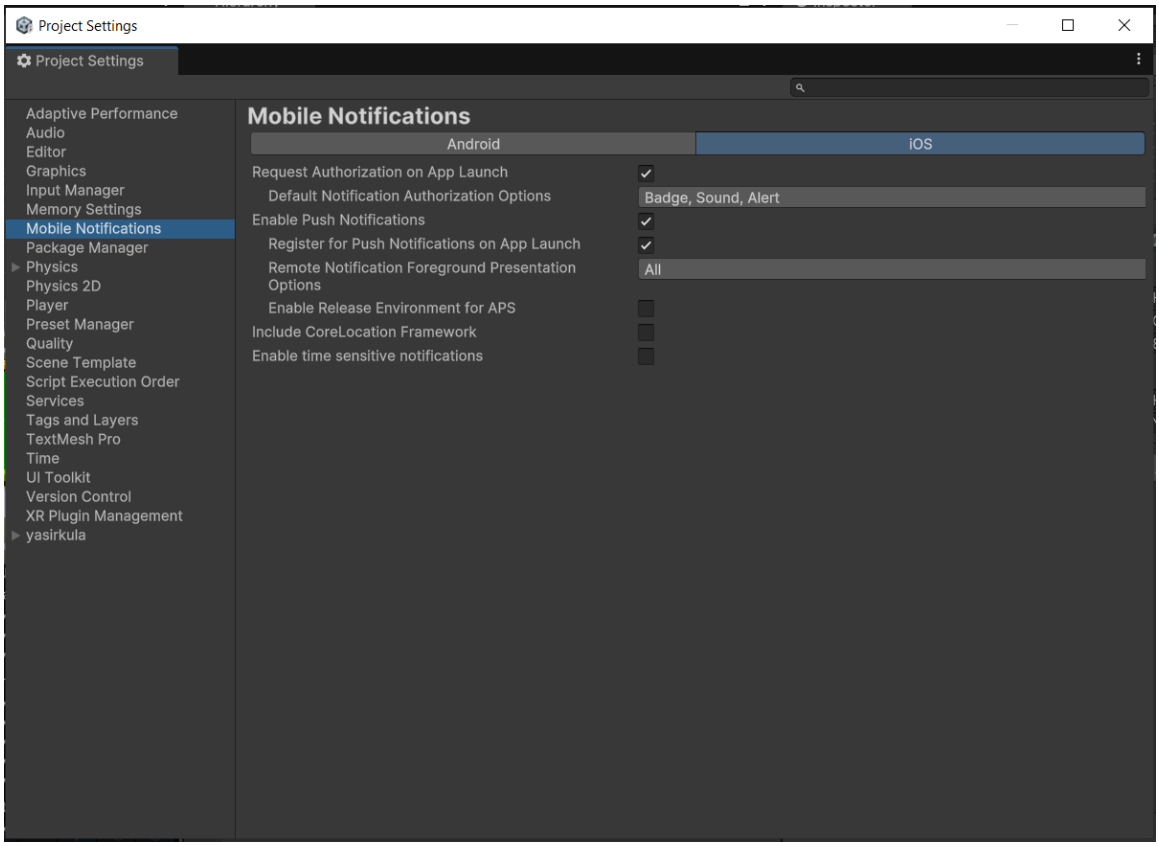
Reward Coins Notification Configs: this is the notification that fire when a reward is ready to claim.

- *Reward Coins Notification Tile:* the tile of the notification when it fired.
- *Reward Coins Notification Text:* the text of the notification when it fired.

You can set up the notification icon by following these steps.

Go to File -> Project Setting -> Mobile Notification

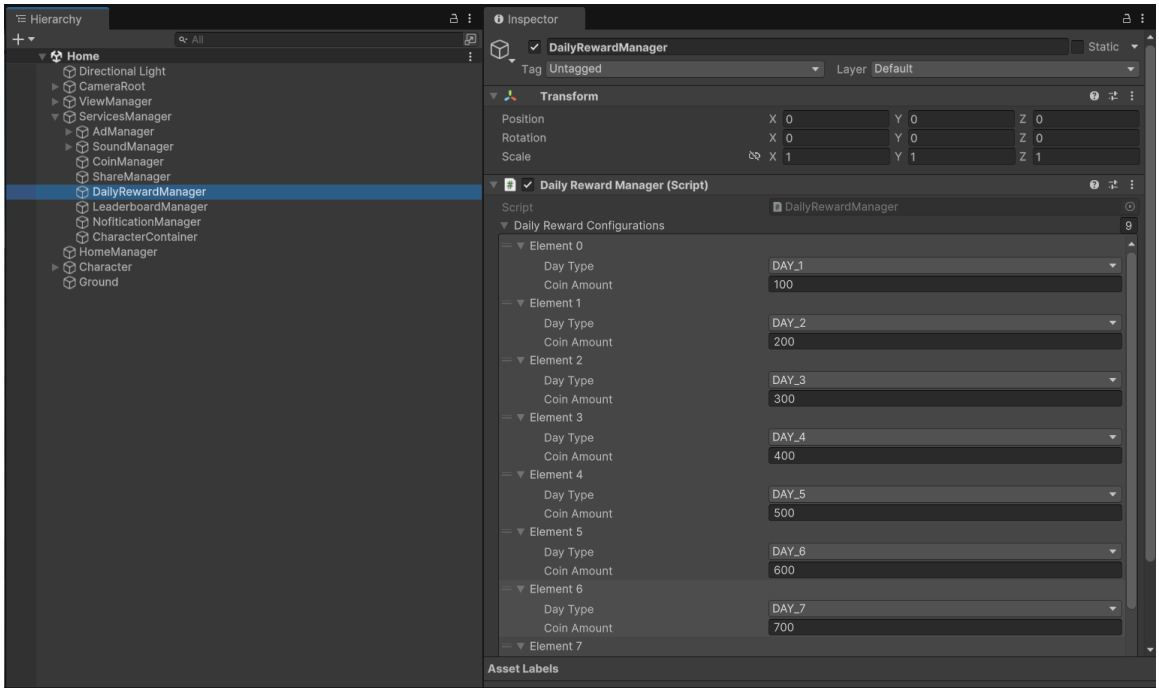




As you can see, there's a filed for you to setup icon for notification. Remember, you must set the Identifier as "icon".

9. Daily Reward Manager

This template has a built-in daily reward system in which the user will be rewarded with coins every predefined interval of time. This is an effective way to increase user engagement and retention for your game. You can configure this feature from the *DailyRewardManager* object in the hierarchy.



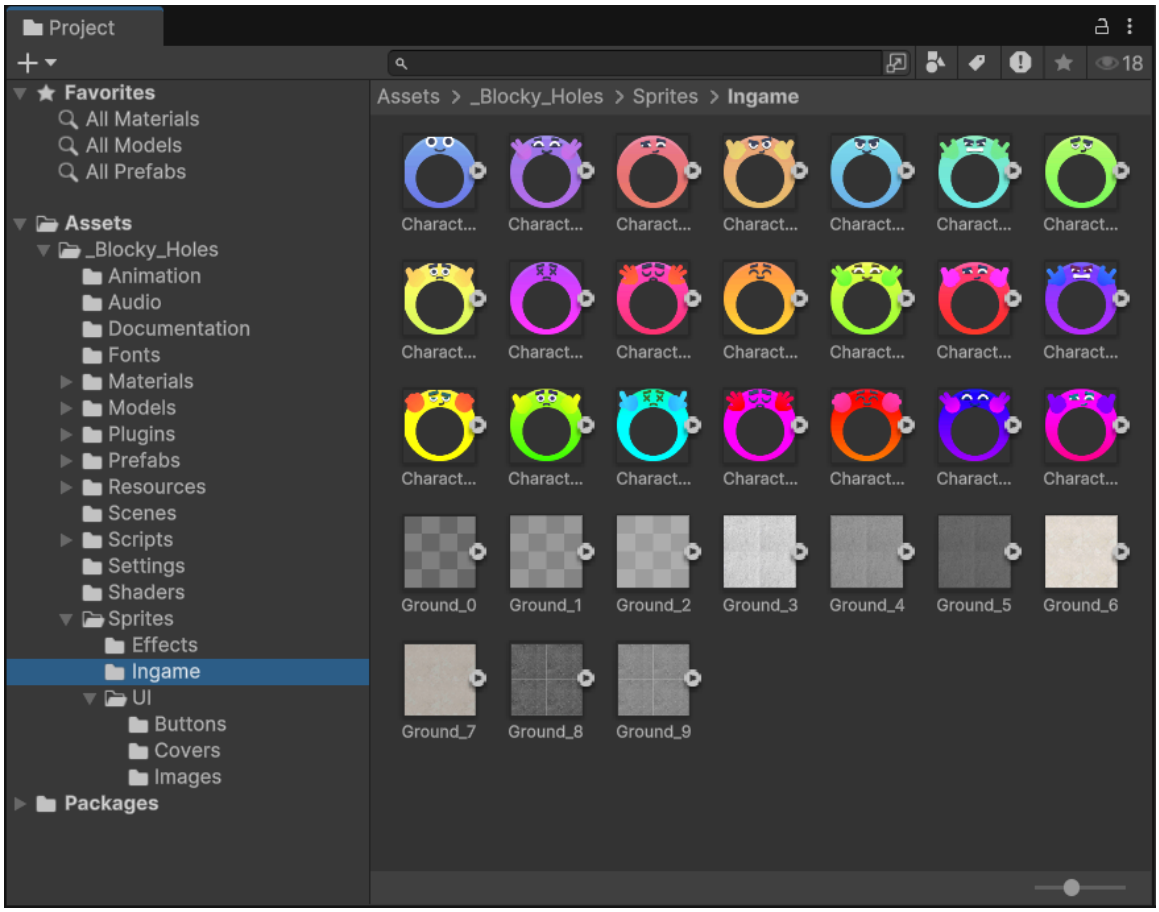
***Daily Reward Configurations**:* this array allows you config the rewarded coin amount for each day.

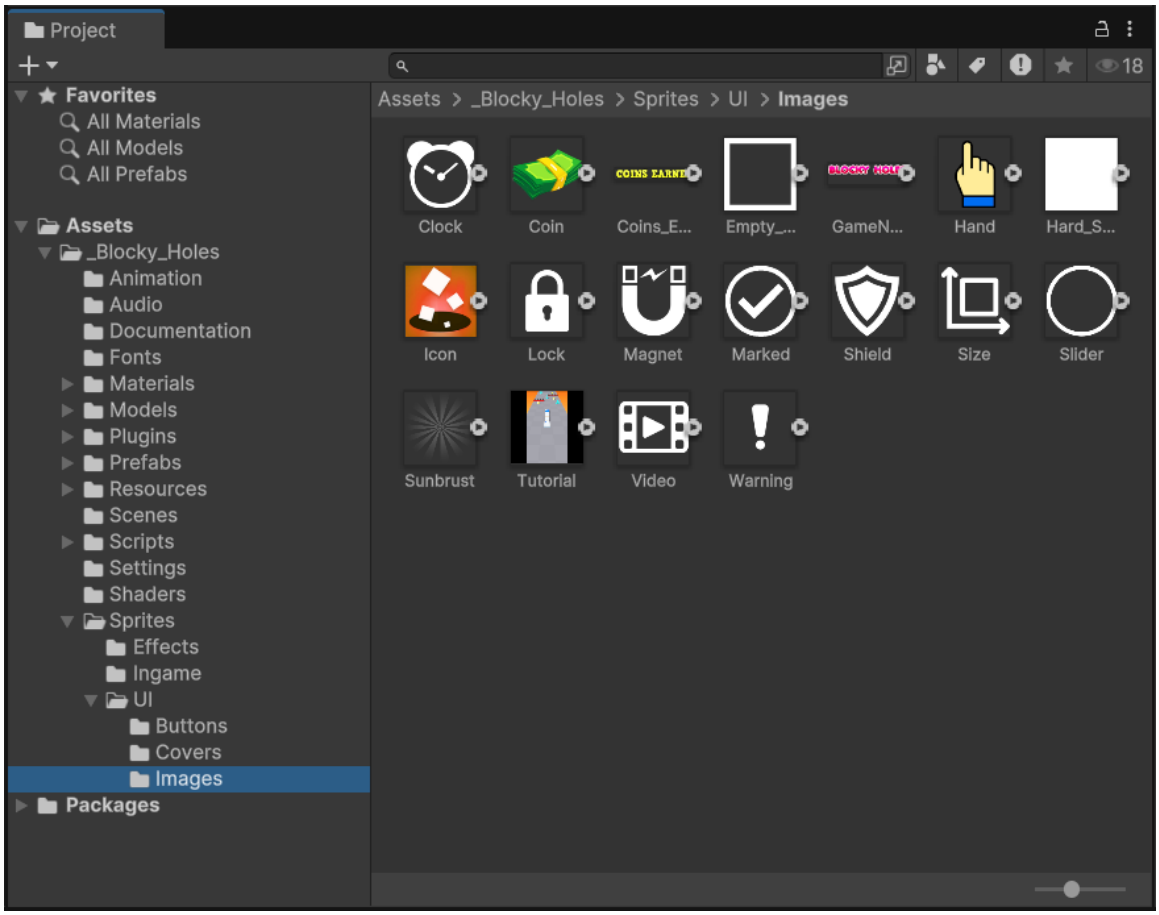
- *Day Type:* the day of the reward.
- *Coin Amount:* the amount of coin player will be rewarded.

***IMPORTANT:** Currently, the daily reward system was designed for only 9 days, after player get all the reward (claimed 9 rewards), the daily reward system will reset to day 1. DO NOT ADD MORE ELEMENT TO THE ARRAY, IT MAY CAUSE UNEXPECTED ERRORS.

10. Customizing UI

All sprites used in this game (for buttons and other UI components) are located under the path *Assets/_Blocky_Holes/Sprites* folder. You can replace them with your own sprites to modify the images as you like.


Project
18
Favorites
All Materials
All Models
All Prefabs
Assets
_Blocky_Holes
Animation
Audio
Documentation
Fonts
Materials
Models
Plugins
Prefabs
Resources
Scenes
Scripts
Settings
Shaders
Sprites
Effects
Ingame
UI
Buttons
Covers
Images
Packages
Assets > _Blocky_Holes > Sprites > Ingame
Charact...
Ground_0
Ground_1
Ground_2
Ground_3
Ground_4
Ground_5
Ground_6
Ground_7
Ground_8
Ground_9


Project
18
Favorites
All Materials
All Models
All Prefabs
Assets
_Blocky_Holes
Animation
Audio
Documentation
Fonts
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Models
Plugins
Prefabs
Resources
Scenes
Scripts
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Shaders
Sprites
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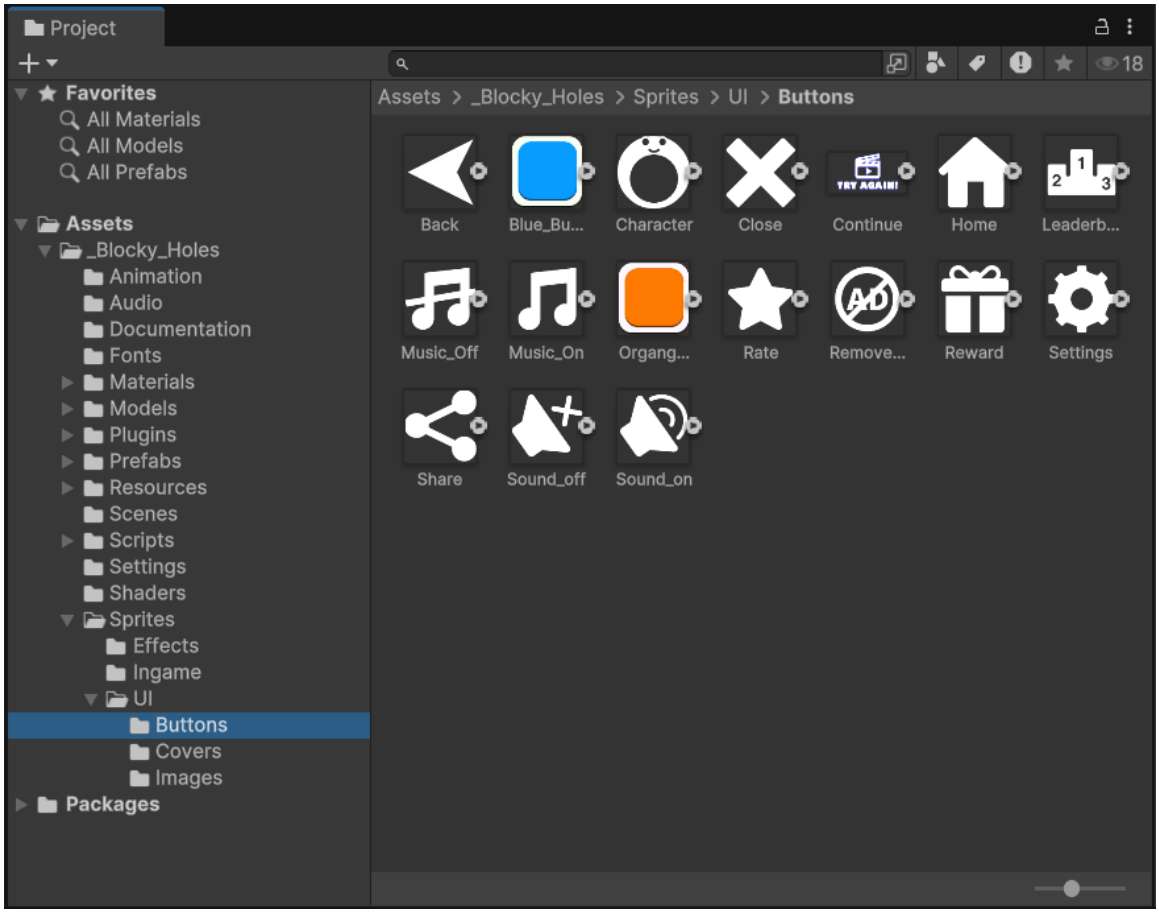
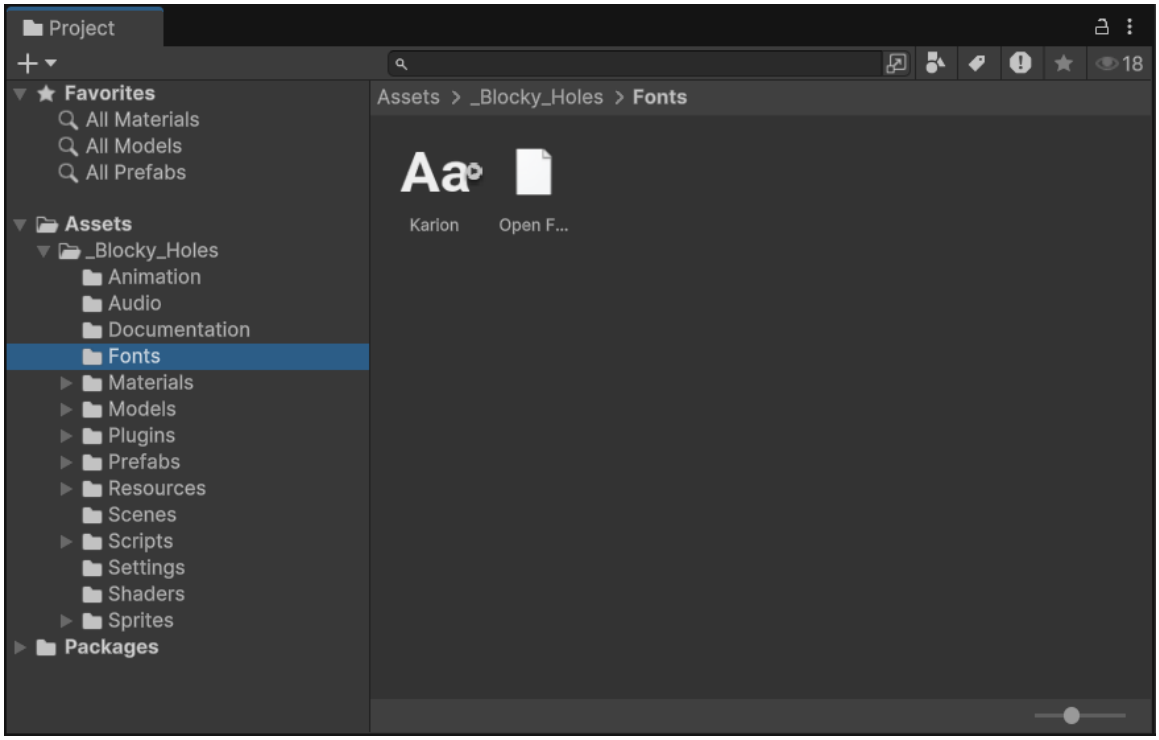
Ingame
UI
Buttons
Covers
Images
Packages
Assets > _Blocky_Holes > Sprites > UI > Images
Clock
Coin
Coins_E...
Empty_...
GameN...
Hand
Hard_S...
Icon
Lock
Magnet
Marked
Shield
Size
Slider
Sunbrust
Tutorial
Video
Warning



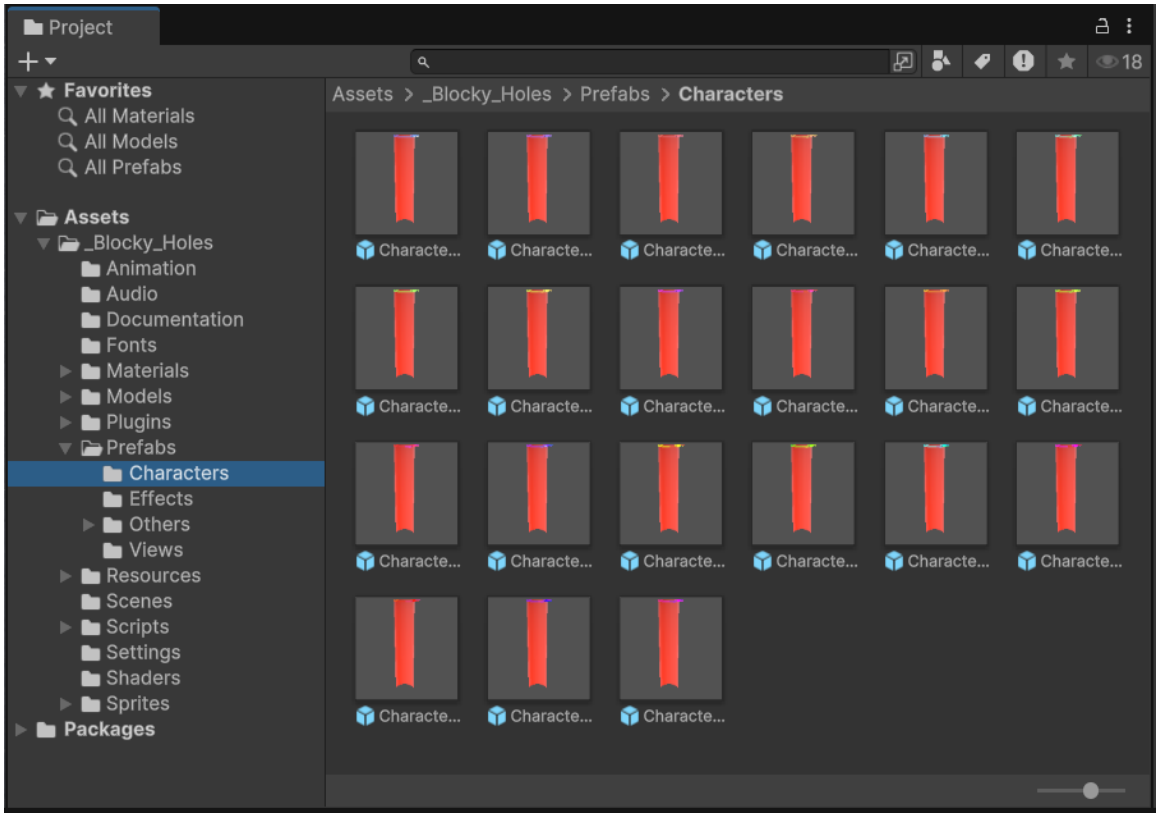
All fonts used in this game are free-to-use in commercial projects. Fonts are located under the path *Assets/_Blocky_Holes/Fonts* folder together with appropriate license files.



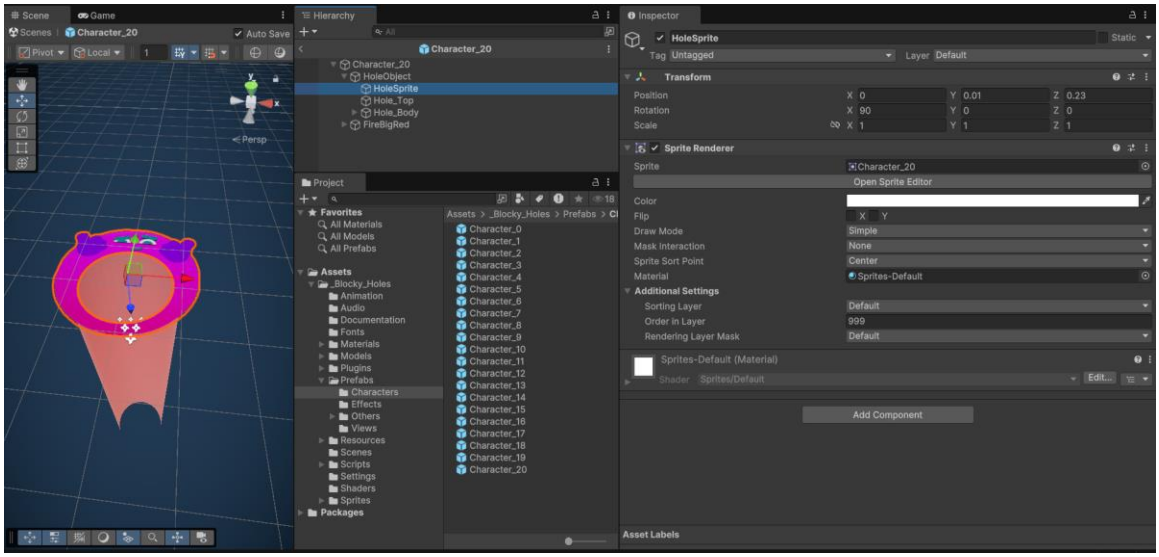
VII. CHARACTER MANAGERMENT

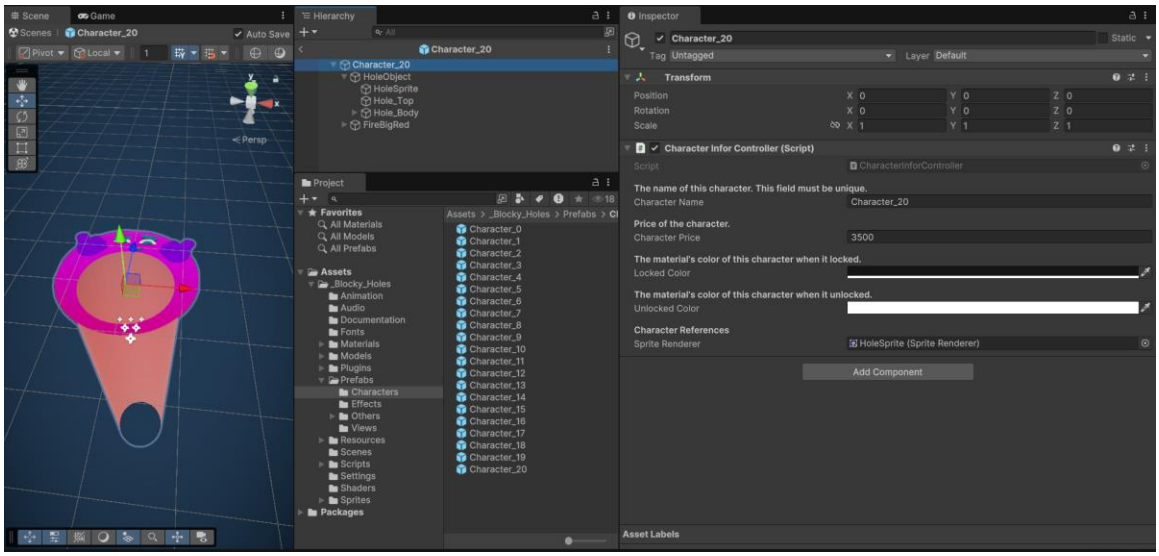
This template comes with built-in 20 characters ready to unlock, each character has only one sprite as the hole. You can find all the characters under the path *Assets/_Blocky_Holes/Prefabs/Characters.* Here's the step to add more characters.

- Create a sprite as the hole. You can use any software out there like paint.net, Photoshop…The software I used to create these models is paint.net.
- Navigate to *Assets/_Blocky_Holes/Prefabs/Characters* and duplicate one of the available character prefabs.

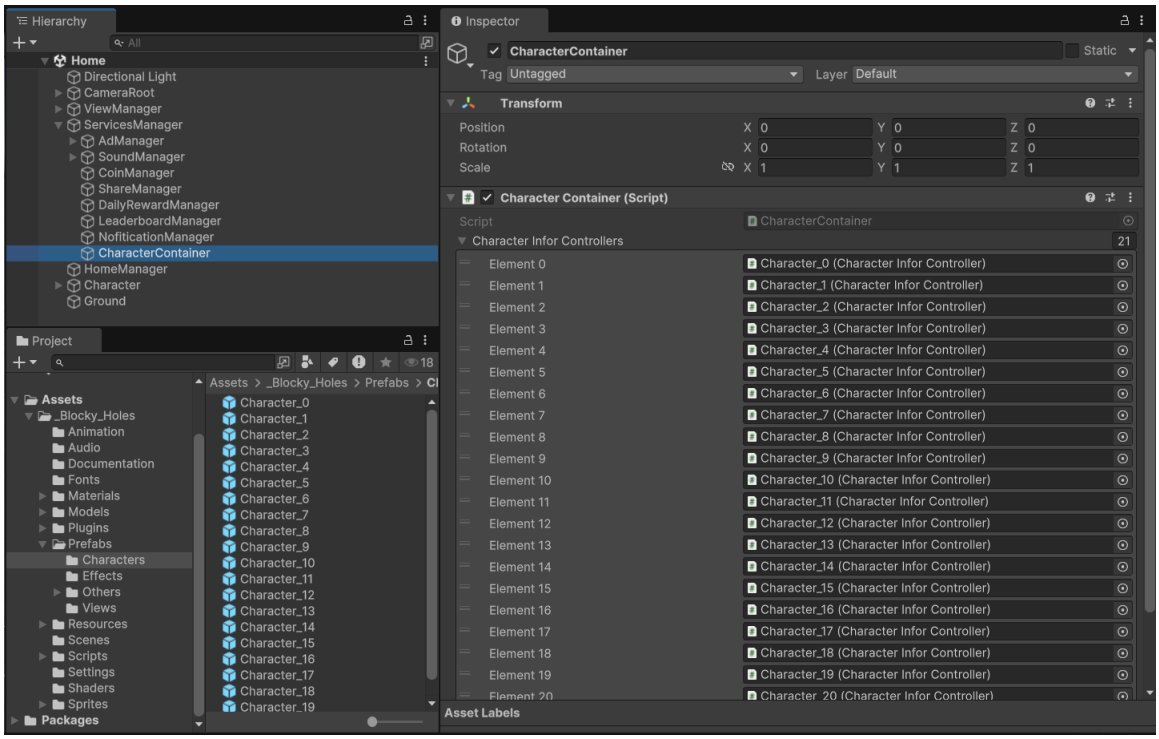


- Change the name of the prefab to a preferred one.
- Replace the *Sprite* in the *Sprite Renderer* component with your new sprite.





- Enter the character name and character price to the *CharacterController* component. If you want to give out this character for free, leave the *Character Price* to be 0 (it will be automatically unlocked). *Locked Color* is the color of the character when it still not un lock. *Unlocked Color* is the color when the character already unlocked. You can change these two colors as you want.
- Resize the character array in *CharacterContainer* game object then drag the new character to it and hit *Ctrl + Z* to save changes.



Now the new character has been added and ready to use in game! You will see it listed in the *Character* scene.

* ***IMPORTANT*:** the new character's name must not repeat any existing character name.

THANK YOU AND GOOD LUCK WITH YOUR GAMES!